Università degli Studi di Napoli Federico II
Scuola Politecnica delle Scienze di Base

Corso di Laurea in

Ingegneria dell'Automazione e Robotica

Dipartimento di Ingegneria Elettrica e delle Tecnologie dell'Informazione

Tesi di Laurea Magistrale

Design and Experimental Evaluation of a ROS-Based Social Robot for Multimodal Mood and Health Monitoring in Office Environments

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Academic Year
2024-2025

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Abstract

Modern office environments often expose young professionals to significant levels of stress, anxiety, and mental health challenges. Despite growing awareness, many individuals remain unaware of their own psychological and physical conditions, leading to decreased well-being and productivity.

This thesis addresses this issue by designing and experimentally evaluating a social robot capable of multimodal mood and health monitoring in office settings. The robot integrates diverse sensing modalities—including facial expression recognition, speech emotion recognition, thermal imaging, and mmWave radar—to estimate users' emotional states and physiological indicators. Importantly, the robot goes beyond passive monitoring by delivering personalized interventions aimed at improving mood, raising health awareness, and fostering a more supportive workplace atmosphere.

We present the hardware and software architecture of the robot, describe the data fusion techniques employed to derive meaningful estimates of mood and health, and detail the interaction strategies designed to influence users positively. Experiments conducted in a living lab environment involve participants performing daily office tasks, during which the robot monitors their mood and health and adapts its behavior in real time.

Statistical analyses, including paired t-tests, demonstrate the potential impact of robot interventions on users' emotional well-being. The results suggest that a multimodal, ROS-based social robot could become an innovative tool for mental health support and awareness in corporate environments, offering new perspectives on the integration of robotics into daily human activities.

Chapter 1
Introduction

Modern office environments have undergone profound transformations in recent decades, driven by technological advances, globalization, and societal changes. However, they remain demanding ecosystems where employees are exposed to intense workloads, high performance expectations, and social pressures. Young professionals, in particular, often experience elevated levels of stress, anxiety, and other mental health challenges as they enter the workforce and attempt to adapt to traditional corporate cultures that can be rigid, hierarchical, and demanding [1].

Recent data from the World Health Organization indicate that one in seven adolescents worldwide suffers from a mental disorder, with suicide ranking as the third leading cause of death among individuals aged 15-29 [2]. As these young people transition into professional life, they frequently carry vulnerabilities that can be exacerbated by the work environment. In the United States, more than half of young professionals aged approximately 22-28 reported needing help for emotional or mental health issues in the past year, while 38% believed that their work environment negatively affected their mental health [1].

These trends are mirrored—and in some aspects, intensified—in Japan. The unique cultural context and strong work ethic in Japanese society impose additional pressures on young workers. A 2023 survey revealed that employees under the age of 30 were the demographic most affected by work-related mental health issues, with 43.9% reporting psychological distress linked to their jobs, up significantly from 29.0% in 2021 [3]. Even workers in their thirties reported increased mental health challenges, rising to 26.8% in 2023 [3]. This increase is partly attributed to the post-pandemic shift back to in-person work, which posed significant adaptation challenges for young employees who began their careers remotely during COVID-19 and had not yet established robust social networks or routines in traditional office settings [3].

In response to longstanding problems such as *karōshi* (death from overwork) and *karōjisatsu* (work-related suicide), the Japanese government implemented a nationwide mandatory “stress check” system in 2015. All companies with more than 50 employees are required to conduct annual standardized surveys to detect early signs of psychological distress, encouraging workers with high stress scores to seek medical consultation [4]. Despite these measures, work-related suicides in Japan remain alarmingly high, with nearly 2,000 individuals taking their lives for job-related reasons in 2021 alone.

Beyond the statistics, cultural factors play a crucial role in how stress and mental health issues manifest and are addressed in the Japanese workplace. A fundamental concept in Japanese culture is “reading the room,” or *kuuki wo yomu* (空気を読む), which refers to the subtle art of sensing unspoken social cues and adjusting one’s behavior to maintain group harmony [5]. This cultural expectation can inhibit individuals from openly expressing distress or seeking help, as there is significant social pressure to maintain composure and avoid burdening others. Employees are expected to intuit the emotional atmosphere, responding with discretion to preserve the collective *wa* (harmony) [5]. While this skill fosters smooth social interactions, it also creates an environment where mental health struggles may remain hidden until they become severe.

Traditional corporate well-being programs often rely on self-reported questionnaires and occasional surveys. However, these tools are inherently subjective and fail to provide continuous or objective monitoring of employees’ emotional and physiological states. This limitation leaves companies unable to detect emerging issues early or to implement timely interventions [1].

Emerging technologies in affective computing and human-robot interaction (HRI) present promising alternatives to traditional approaches.

Robots equipped with multimodal sensors can analyze diverse human signals—facial expressions, speech prosody, thermal signatures, and physiological data—to estimate emotional states and health indicators in real-time

[6, 7]. Notably, studies have demonstrated that social robots can effectively promote mental well-being in workplace settings. For example, research at the University of Cambridge showed that robots acting as "well-being coaches" were able to deliver brief interventions promoting positive psychology practices such as gratitude and recall of positive experiences. Participants reported that interactions with these robots contributed to improved mental well-being, although perceptions varied depending on the robot's appearance [8].

In Japan, similar initiatives have emerged. Fuji Xerox, in collaboration with the University of New South Wales, has developed a prototype social robot designed to integrate seamlessly into office environments and monitor employees' well-being while encouraging positive interpersonal connections. These robots are envisioned as "empathetic companions" capable of subtly detecting emotional tension and acting as mediators to foster a harmonious workplace atmosphere.

Technologically, several modalities contribute to the robot's ability to assess human emotional and physical states. Facial expression recognition (FER) analyzes micro-expressions and facial action units to classify emotional states, achieving high accuracy in controlled environments [9]. Speech emotion recognition (SER) exploits features such as pitch, prosody, and spectral characteristics to detect emotions like stress, sadness, or happiness [10]. Thermal imaging can detect subtle temperature changes in facial regions associated with vasomotor responses to emotional states—e.g., nasal tip cooling during stress or anxiety [11]. Additionally, millimeter-wave (mmWave) radar sensors can non-invasively measure vital signs like heart rate and respiration, which fluctuate with emotional arousal [12].

The potential of integrating these technologies aligns closely with Japan's ambitious Moonshot R&D Program. Launched in 2020 by the Japan Science and Technology Agency, the Moonshot Program envisions creating societies where humans and robots coexist harmoniously and collaboratively, improving quality of life and well-being. Specifically, Moonshot Goal 3 aims to develop AI-driven robots capable of autonomous learning, environmental adaptation, and evolving intelligence, capable of working side-by-side with humans by 2050 [13]. This vision includes social robots capable of perceiving and responding to complex human emotions, seamlessly integrating into workplaces and supporting both physical tasks and mental health interventions.

This thesis explores the design and experimental validation of a ROS-based social robot tailored for multimodal mood and health monitoring in office environments. Our robot integrates data streams from facial expression recognition, speech emotion recognition, thermal imaging, and mmWave radar sensing. The novelty of this work lies in a real-time data fusion pipeline that synthesizes these diverse inputs to estimate users' emotional and physiological states with practical reliability. Beyond passive monitoring, the robot delivers brief, personalized interventions intended to improve users' mood, raise awareness of potential health issues, and contribute to a healthier and more harmonious workplace atmosphere. These interventions are designed to align with cultural norms such as the *kuuki wo yomu* introduced earlier, ensuring that the robot's actions are contextually appropriate and respectful of personal boundaries.

Problem Statement and Research Hypotheses. This thesis investigates whether a multimodal, contactless robotic system can (i) reliably estimate users' mood and health in real time and (ii) positively influence emotional well-being through adaptive interventions in office environments. Accordingly, the research is guided by the following hypotheses:

1. **H1:** Multimodal fusion of facial expression, speech emotion, thermal, and mmWave signals can approximate human assessments of affective state with acceptable reliability.

2. **H2:** Robot-led interventions have a measurable, immediate positive effect on users' reported mood after the interaction.

3. **H3:** Supplying the dialogue system with fused sensing context (FER, temperature, identity) enhances engagement and improves outcomes compared to context-free interactions.

The remainder of this thesis is organized as follows. Chapter 2 reviews relevant literature and related works in social robotics, affective computing, and health monitoring technologies. Chapter 3 outlines the specific research goals of this project. Chapter 4 describes the robot's hardware components, while Chapter 5 details the software architecture and data processing pipelines. Chapter 6 presents the robot's intervention strategies. Chapter 7 discusses our experimental methodology and results. Finally, Chapter 8 concludes the thesis and proposes directions for future research and development.

Chapter 2

Background and Related Works

2.1 Introduction to Social Robots in Workplaces

Social robots are physical entities designed to engage in human-like interactions through verbal and non-verbal communication [14]. Unlike purely software-based systems, social robots leverage their physical embodiment to foster trust, empathy, and engagement with users. This makes them uniquely suited for applications in professional environments where human relationships and social dynamics are crucial.
In modern workplaces, increasing stress levels, coupled with the demand for high productivity and psychological well-being, have motivated researchers to explore the potential of social robots as supportive tools. They are envisioned not merely as task executors but as companions, coaches, and mediators who can enhance both individual and group well-being.

2.2 Social Robots for Workplace Well-Being

An emerging line of research investigates how social robots can act as well-being coaches, delivering interventions aimed at stress reduction and promoting positive mental health practices in professional environments.
Spitale et al. [8] conducted an in-the-wild study in a technology company, where two robot "well-being coaches" (QTrobot and Misty) delivered weekly positive psychology exercises to employees. The study demonstrated that the robots could effectively engage staff in well-being activities, although the robots' physical form significantly influenced user acceptance and engagement levels.
Similarly, Jeong et al. [15] explored a robotic positive psychology coach

that delivered daily sessions to college students. Although the context was academic rather than corporate, the findings showed improvements in participants' self-reported well-being, suggesting potential applications in workplace settings.

Robinson et al. [16] carried out a randomized controlled trial in which a humanoid robot led a single-session mindfulness exercise. Participants found the session both feasible and enjoyable, highlighting robots' potential to deliver brief mental health interventions during the workday.

Lopes et al. [17] conducted a longitudinal workplace wellness program where employees received health-promoting interventions from either a human coach or a social robot. The robot-led group achieved outcomes comparable to, and sometimes exceeding, those led by humans, indicating that social robots can serve as credible and effective health-promoting agents in office settings.

Collectively, these studies underscore the feasibility and potential benefits of deploying social robots as well-being coaches in professional environments. However, most implementations remain experimental, often with limited deployment duration and scale.

2.3 Social Robots for Team Dynamics and Social Mediation

Beyond individual well-being, researchers have explored how social robots can influence group dynamics and interpersonal relationships among coworkers.

Sebo et al. [18] demonstrated that a robot teammate who displayed vulnerability—such as admitting mistakes—could positively impact human team members' trust and empathy, thereby enhancing overall team cohesion. Such findings suggest that robots can serve as social catalysts, fostering psychological safety within teams.

Conversely, Savela et al. [19] reported that merely introducing a robot as a team member could reduce human team members' identification with their group. This "us vs. them" effect highlights the risk that robots might inadvertently weaken team cohesion if not carefully integrated into workplace dynamics.

Weisswange et al. [20] provided a comprehensive scoping review of how robots can act as mediators in group settings. Their work identifies multiple strategies through which robots can improve group interactions, such as enforcing turn-taking, resolving conflicts, or injecting humor to diffuse tension. These studies collectively emphasize that while robots hold promise for enhancing team dynamics, successful deployment requires thoughtful design to ensure robots act as facilitators rather than disruptors of workplace harmony.

2.4 Social Robots for Emotion and Physiological Monitoring

A critical aspect of supporting well-being in workplaces is the ability to monitor employees' emotional and physiological states unobtrusively and in real time. Social robots, equipped with multimodal sensing technologies, offer new possibilities in this domain.

Klęczek et al. [21] investigated a humanoid robot guiding individuals through a stress-reduction breathing exercise while simultaneously recording physiological signals, such as EEG and skin conductance. The study demonstrated that the robot's presence could influence emotional arousal and provided real-time feedback on users' stress levels.

Bieber et al. (2018) [22] demonstrated the feasibility of touchless heart rate recognition using robot-mounted cameras and photoplethysmography algorithms. Such capabilities could allow robots to assess employees' physiological stress levels discreetly during routine office interactions.

These technologies, while still largely experimental, underscore the potential for social robots to perform real-time emotional and health monitoring in workplaces. However, challenges remain regarding privacy, data security, and user acceptance.

2.5 Culturally Sensitive Human-Robot Interaction

Designing social robots for workplace deployment necessitates sensitivity to cultural norms and communication styles, particularly in high-context cultures like Japan.

Lim et al. [23] reviewed how cultural factors shape human-robot interaction. They concluded that robots aligning with local social norms, language styles, and gestures are more likely to be accepted and trusted by users. For instance, in Japan, politeness, indirect communication, and sensitivity to group dynamics are highly valued, suggesting robots must embody these traits to integrate effectively into Japanese workplaces.

Jung [5] explains the Japanese concept of "reading the air" (*kuuki wo yomu*), which involves perceiving unspoken social cues and adjusting one's behavior to maintain harmony. Social robots intended for Japanese offices may need to exhibit similar awareness, adapting their interactions based on subtle changes in the emotional atmosphere.

These cultural considerations are vital, as a robot perceived as culturally insensitive could quickly lose user trust and become counterproductive in fostering well-being or teamwork.

2.6 Gap and Motivation for This Work

Although substantial research has explored social robots for well-being coaching, team dynamics, and even emotion monitoring, significant gaps remain. Most existing systems:

- Rely on single-modality sensing (e.g., only facial expressions or speech) rather than integrating multiple data streams for comprehensive assessment.
- Operate in controlled experimental conditions rather than continuous, real-world office environments.

- Often involve direct physical contact or invasive sensors such as EEG, which may not be suitable or acceptable in everyday office settings.
- Focus predominantly on passive monitoring rather than proactive interventions tailored to users' current emotional and physiological states.
- Lack cultural adaptation necessary for deployment in contexts like Japanese workplaces.

This thesis aims to bridge these gaps by designing and experimentally validating a ROS-based social robot capable of:

- Real-time fusion of multimodal sensory data—including facial expression recognition, speech emotion recognition, thermal imaging (for fever detection), and mmWave radar sensing.
- Estimating both mood and physical health indicators in an unobtrusive, privacy-respecting manner.
- Delivering personalized interventions aimed at improving employees' emotional well-being and raising health awareness.
- Operating in a culturally sensitive way, aligning robot behaviors with workplace norms and practices, particularly in Japanese contexts.

By integrating these components, this work represents a novel step toward deploying social robots as effective tools for supporting mental health and well-being in modern office environments, while ensuring comfort, privacy, and minimal intrusion for users.

Chapter 3

Research Goal

3.1 Methodological Framework

The research presented in this thesis adheres to rigorous experimental principles as outlined in the guide by Hoffman and Zhao [24]. Their framework provides a structured methodology for conducting experiments in Human-Robot Interaction (HRI), including recommendations on experimental design, formulation of research questions, hypothesis definition, statistical analysis, and transparent reporting.
In alignment with these guidelines, this thesis ensures:

- Clear formulation of research questions prior to experimentation, avoiding post-hoc hypothesis formulation (HARKing).
- Definition of both subjective measures (e.g., user perception of well-being, robot likability) and objective measures (e.g., sensor data, physiological signals).
- Consideration of experimental design choices (e.g., between-subjects, within-subjects, or mixed designs).
- Planning for appropriate sample sizes and statistical power.
- Transparent reporting of results, including effect sizes and statistical significance.
- Documentation of limitations and considerations for replicability.

By following this methodological framework, the research aims to contribute reproducible and scientifically sound findings to the field of HRI.

3.2 Motivation

Social robots hold significant potential for promoting mental health and well-being in workplace environments. However, as discussed in the previous chapter, current research often focuses on isolated sensing modalities, short-term experimental studies, and contexts that differ substantially from everyday office dynamics. Furthermore, many systems rely on invasive technologies, such as EEG or wearable sensors, which may not be practical or culturally acceptable in professional environments.

3.3 Research Objectives

The primary objectives of this research are:

1. **Design and Implementation of a ROS-Based Social Robot System:** Develop a robotic platform equipped with multimodal sensors capable of operating in office environments.

2. **Multimodal Data Fusion:** Implement data fusion algorithms, in order to produce reliable estimations of users' emotional and physical health states, gathering information from:

   - Facial Expression Recognition (FER)
   - Speech Emotion Recognition (SER)
   - Thermal imaging (limited to fever detection, not emotion recognition)
   - mmWave radar sensing

3. **Real-Time Monitoring:** Develop software pipelines that enable real-time acquisition, processing, and interpretation of multimodal data without disrupting normal office activities.

4. **Intervention Strategies:** Design intervention protocols that the robot can execute to influence users' mood positively or provide health-related guidance.

5. **Experimental Validation:** Conduct experiments in a living lab office environment to evaluate:
    - The accuracy of mood and health estimations.
    - The effectiveness of robot-led interventions.
    - User acceptance and perceived helpfulness of the robotic system.

3.4 Research Questions

In line with best practices suggested by Hoffman and Zhao [24], this thesis explicitly defines research questions prior to experimentation to ensure clarity and avoid retrospective hypothesis formulation. The following research questions guide this work:

- **RQ1:** Can multimodal non-invasive sensing technologies, when integrated into a unified system, reliably estimate employees' mood and physical health in real time within an office environment?
- **RQ2:** Are robot-led interventions effective in improving employees' mood or raising health awareness during typical office activities?

3.5 Contribution Statement

Through the design, implementation, and evaluation of a ROS-based social robot system, this thesis contributes:

- A technical framework for fusing multimodal, non-invasive sensory data for mood and health estimation.
- A practical demonstration of real-time robot-led interventions in a workplace setting.

By addressing these objectives and following a structured experimental methodology, this work aims to bridge a critical gap between laboratory research and practical deployment of social robots as supportive tools for mental health and well-being in modern office environments.

Chapter 4

Hardware

This chapter describes the hardware architecture of SUNNY, the custom-built social robot developed for this research. The design combines a compact, mobile base with a powerful computational platform and a diverse set of sensors and output devices to support multimodal perception and interaction in office environments.

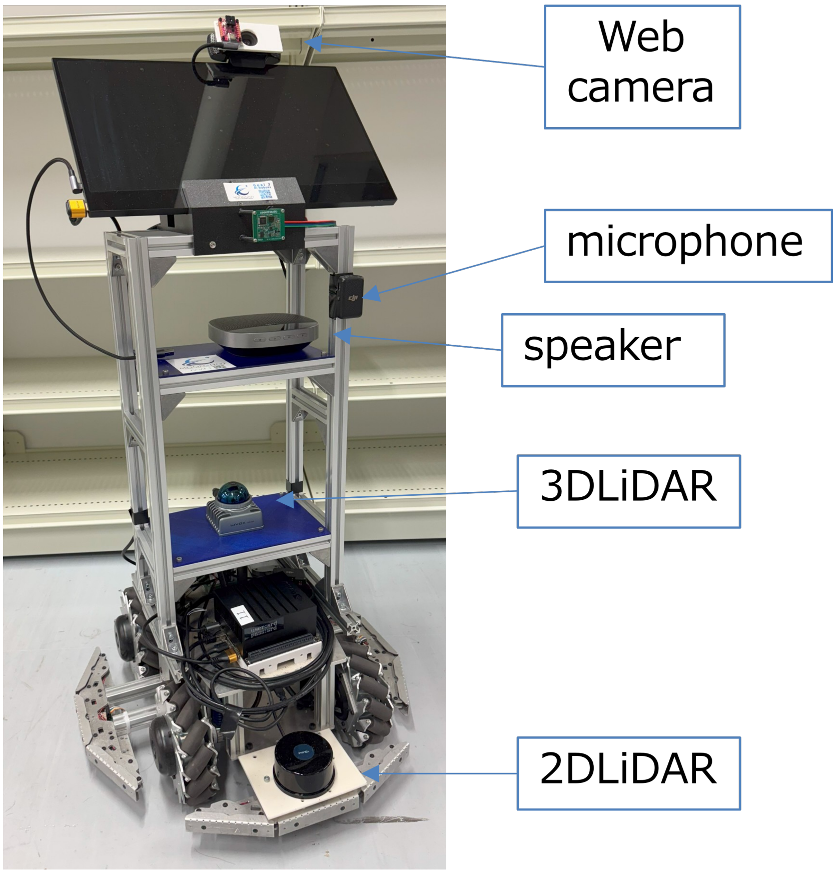

Figure 4.1: Robot Physical Structure and Components

4.1 Robot Physical Structure

SUNNY is built on the Vstone Wheel Robot ver. 3.0 mobile base, which uses mecanum wheels to enable omnidirectional movement. This capability allows the robot to navigate confined office spaces with precise lateral, diagonal, and rotational motions. The robot's structure is composed of aluminum bars and various custom components designed and fabricated using 3D printing with PVC materials. This combination ensures mechanical rigidity while maintaining low weight and modularity for easy hardware modifications.

4.2 Computing Unit

The robot's computational core is the NVIDIA Jetson AGX Xavier Developer Kit, chosen for its high-performance GPU and efficient AI processing capabilities. This platform enables real-time inference for vision-based tasks, speech analysis, and sensor data fusion. The system operates on Ubuntu 20.04 LTS with ROS Noetic as the robotics middleware, allowing seamless integration of hardware drivers and high-level robot functionalities.

4.3 Sensors and Input Devices

SUNNY integrates multiple sensors to perceive its environment and users:

- **RGB Camera:** A Logitech C920 webcam captures high-resolution video streams used for facial expression recognition (FER).

- **Thermal Camera:** A FLIR Lepton 3.5 thermal sensor combined with the PureThermal 3 interface board measures facial surface temperature.

- **mmWave Radar Sensor:** The Seeed Studio R60A mmWave radar operates at 60 GHz to non-invasively measure physiological parameters such as heart rate and respiration, enabling health monitoring without physical contact.

- **Microphone Array:** Audio input is acquired through either the Anker PowerConf S500 speakerphone or the DJI Mic system. Notably, the DJI Mic was used during demonstrations at ICRA 2024 in Yokohama due to its superior performance in high-noise environments.

- **2D LiDAR:** The YDLIDAR TG30 laser rangefinder is installed near the robot's base to detect obstacles close to the floor. It provides 360-degree scans up to 12 meters with a frequency of up to 12 Hz, ensuring effective obstacle avoidance for low-level objects.

- **3D LiDAR:** The Livox Mid-360 sensor, mounted near the robot's center of mass, offers a 360-degree horizontal field of view and a vertical field of view of 59 degrees. This sensor enables volumetric mapping and detection of objects and people that are not at floor level. In later chapters, it will be explained how data from both LIDARs are fused to create a unified obstacle avoidance system.

4.4 Output Devices

The robot's interactive capabilities are realized through several output devices:

- **Display:** A generic 1920x1080 LCD touchscreen display is mounted on the robot to show visual feedback, icons, or text during interactions. While the touchscreen functionality is available, it was not utilized in the current implementation but will be considered for future developments.

- **Speakers:** Audio output is handled by the Anker PowerConf S500 speakerphone, enabling the robot to provide spoken feedback and engage in dialogue during interventions.

- **Mobile Base:** The Vstone mecanum-wheel base allows precise movements in any direction, contributing to the robot's interactive presence and enabling safe navigation in dynamic office spaces.

4.5 Power and Battery

SUNNY is powered by an internal battery system, providing approximately 2 hours of operational autonomy under typical usage conditions. Future developments include plans to integrate an autonomous docking system with rapid charging capabilities, allowing the robot to automatically return to a charging station when its battery levels are low.

4.6 Hardware Architecture Diagram

Figure 4.2 illustrates the block diagram of SUNNY's hardware architecture, highlighting the relationship between sensors, the central computing unit, and output devices.

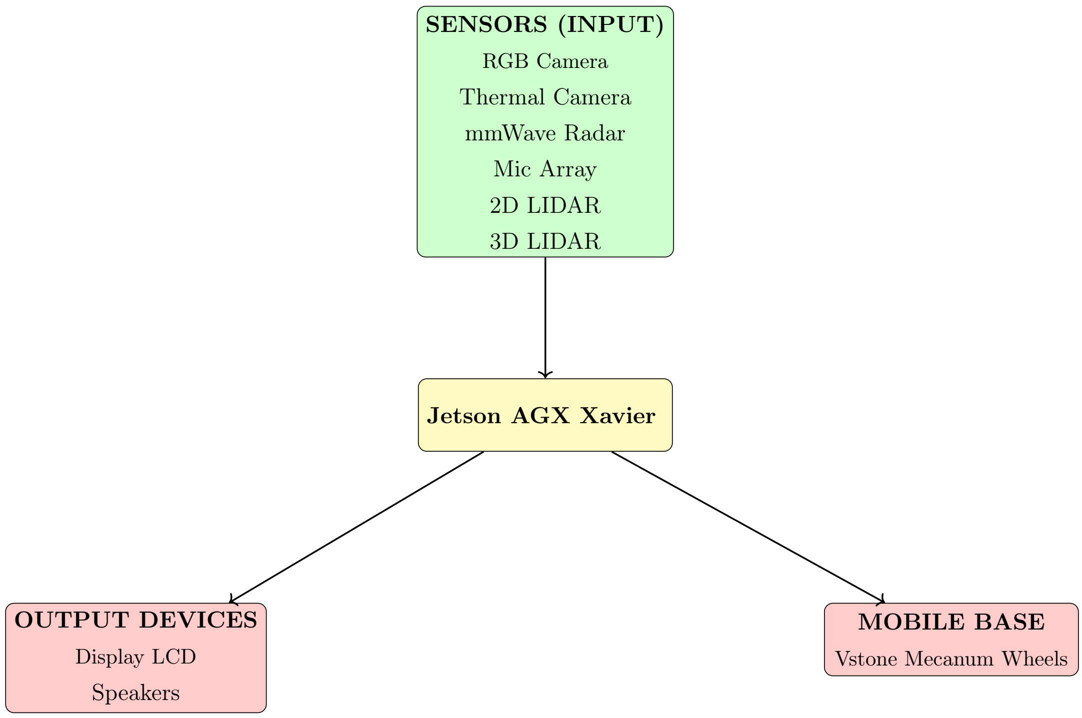

Figure 4.2: Hardware block diagram of the SUNNY robot.

The chosen hardware configuration provides the necessary computational power and sensory capabilities for real-time, multimodal perception and interaction, forming the foundation for the robot's role in monitoring and enhancing workplace well-being.

Chapter 5

Software Architecture

5.1 Overview

The software architecture of the SUNNY robot has been designed to support real-time multimodal sensing, context-aware interaction, and autonomous navigation in dynamic office environments. The system is built on top of the Robot Operating System (ROS) Noetic framework, which provides modularity, interoperability, and extensive support for sensor integration and robot control.

The overall architecture follows a node-based paradigm in which each functional component—such as facial expression recognition (FER), speech emotion recognition (SER), thermal and mmWave sensing, interaction management, and navigation—is encapsulated in a dedicated ROS node. These nodes communicate via publish/subscribe and service mechanisms over defined topics, enabling flexible data flow and scalability.

The system runs on Ubuntu 20.04 and is deployed on an NVIDIA Jetson AGX Xavier Developer Kit. This platform supports CUDA acceleration and efficient deep learning inference at the edge, which is essential for running computer vision and emotion recognition models in real-time.

The software is organized into four main layers:

1. **Perception Layer** – acquires and processes raw data from onboard sensors (camera, thermal, microphone, mmWave radar, LIDAR).
2. **Estimation Layer** – fuses the sensor inputs to estimate user mood and health parameters.
3. **Interaction Layer** – determines and triggers appropriate robot behaviors based on the estimated state.

4. **Navigation Layer** – enables autonomous movement through SLAM and obstacle avoidance based on 2D and 3D LIDAR data.

In the following sections, each software module is described in detail, including its internal logic, interaction with other nodes, and role within the complete system.

5.2 Sensing Modules

The effectiveness of a social robot in assessing user mood and health relies critically on its sensing capabilities. To this end, SUNNY integrates a diverse set of non-invasive sensors designed to capture multimodal information—including visual cues, audio signals, thermal data, and physiological metrics. Each sensor is associated with a dedicated ROS node that handles data acquisition, preprocessing, and communication with the rest of the system.

The decision to use only contactless sensors was driven by both technical and ethical motivations. In contrast to approaches relying on wearable devices or EEG-based systems, our architecture preserves user comfort and privacy while enabling continuous monitoring in open office environments.

This section provides a detailed description of each sensor module used in SUNNY, covering its hardware configuration, software implementation, data processing pipeline, and integration into the broader ROS ecosystem. Specifically, we describe the modules for:

- Facial Expression Recognition (FER), based on RGB camera input and lightweight DNN models;
- Speech Emotion Recognition (SER), using transformer-based audio analysis;
- Thermal sensing for non-contact temperature estimation;
- mmWave radar for vital sign monitoring such as heart rate and breathing.

Each module contributes a unique modality to the multimodal fusion framework, enabling a robust and privacy-aware estimation of the user's emotional and physiological state.

5.2.1 Facial Expression Recognition (FER)

To detect users' facial expressions in real-time, we developed a ROS node written in C++, based on OpenCV DNN modules and integrated with ROS Noetic. The system is optimized for the Jetson AGX Xavier platform by leveraging CUDA acceleration.

Pipeline and Models

The FER pipeline includes three main steps:

- **Face Detection:** We use *YuNet*, a lightweight and efficient face detector developed by Intel, capable of real-time detection in embedded environments [25].

- **Face Recognition:** To personalize robot interaction while preserving user privacy, we implemented a lightweight face recognition module based on the *SFace* model [26]. Unlike traditional biometric systems that require cloud-based identification or large datasets, our approach uses local image features extracted during an enrollment phase with prior consent.
  Participants in the study (e.g., lab members) were asked to provide a single frontal image of their face. These images were stored locally in a designated folder, with filenames encoding the user's *name* and *preferred language*. Each image was processed offline by a C++ program built using OpenCV's DNN module. The system performed face detection using **YuNet** (as told before), followed by face alignment and embedding generation using **SFace**. For each image, a 512-dimensional feature vector was extracted and saved in a local '.txt' file, alongside the corresponding filename identifier.

During live operation, the robot uses a Logitech C920 camera to capture real-time RGB frames. Each incoming frame is processed by the FER node, which detects the face and computes its SFace embedding in real time. The resulting vector is then compared to all the reference vectors stored in a '.txt' file using Euclidean distance. If the distance to one of the stored vectors is below a predefined threshold γ (decided through tryal experiences), the system considers it a match and extracts the associated user metadata (name and preferred language) from the matched filename. These details are then published and forwarded to the interaction engine.

- **Facial Expression Classification:** The expression is inferred using a *MobileFaceNet*-based architecture fine-tuned for facial emotion recognition. This model is known for its low computational cost and robustness across multiple facial variations [27].

ROS Integration and Output

The node subscribes to an image stream from a Logitech C920 camera and publishes two outputs:

1. An annotated image stream on the topic `/face_image`, displaying face bounding boxes and predicted emotion labels.
2. A structured message on the topic `/face_expression`, including timestamp, person ID, and detected emotion label (e.g., happy, sad, angry, etc.).

Furthermore, the system logs the detected expressions in CSV format, enabling later analysis and offline evaluation. This multimodal synchronization is critical for integrating FER outputs with those from speech emotion recognition (SER), thermal sensing, and mmWave radar, which together inform the overall mood estimation pipeline.

Privacy Considerations

Unlike other systems that rely on biometric databases or wearable sensors, our FER module is entirely vision-based and designed to operate in compliance with privacy-preserving constraints. No facial images or biometric data are transmitted externally or stored beyond local RAM and the temporary '.txt' files. All comparisons are performed locally on the Jetson Xavier platform. Moreover, SFace itself is designed as a privacy-conscious model that avoids identity leakage by relying on synthetically generated training data via conditional GANs [26]. This architecture allows for basic user recognition and personalized interaction without requiring the use of sensitive or invasive technologies.

5.2.2 Speech Emotion Recognition (SER)

To estimate the user's emotional state from audio input, the system includes a dedicated ROS node that performs Speech Emotion Recognition (SER). The node leverages a pre-trained transformer-based model, specifically the **SER-Odyssey-Baseline-WavLM-Categorical** model [28], fine-tuned on the MSP-Podcast corpus [29]. This model was released as part of the Odyssey 2024 SER Challenge and represents the current state of the art for emotion recognition from naturalistic, non-acted speech.

Architecture and Implementation

The SER module subscribes to a ROS topic `interaction_info`, which provides metadata on the user-robot interaction including timestamps for audio acquisition. Once a new audio file is detected, the system:

1. Loads the .wav file and resamples it to match the model's expected input frequency.
2. Normalizes the waveform using the model's configuration parameters (mean and standard deviation).

3. Processes the audio through the `WavLM-Large` encoder [30], followed by attentive statistics pooling and a fully connected classification head.

4. Applies a softmax activation to obtain emotion probabilities.

5. Selects the most probable emotion from the predefined categorical set: `Angry, Sad, Happy, Surprise, Fear, Disgust, Contempt, Neutral`.

6. Publishes the resulting label on the ROS topic `ser_results`.

Data and Model

The **MSP-Podcast Corpus** is a large-scale dataset containing over 237 hours of spontaneous, emotionally rich speech segments. These audio samples are extracted from online podcasts and annotated for categorical emotions (e.g., happiness, sadness, anger) as well as emotional dimensions such as arousal, valence, and dominance. Each utterance is independently labeled by at least five human annotators to ensure labeling consistency and reliability.

The SER-Odyssey baseline model, which leverages WavLM embeddings for audio classification, achieves a **macro-F1 score** of 0.327 in categorical emotion recognition tasks. The macro-F1 score is a performance metric that considers both precision and recall, averaged across all emotion classes equally—regardless of class imbalance. This makes it particularly suitable for tasks like emotion recognition, where certain emotions (e.g., disgust or fear) may appear much less frequently than others.

Integration with the Robot

The SER node runs in real time on the NVIDIA Jetson AGX Xavier, leveraging PyTorch and the HuggingFace Transformers library. It works in coordination with other perception modules such as facial expression recognition (FER) and thermal sensing, contributing to the multimodal estimation pipeline. The result is used not only for mood tracking but also for dynamic adaptation of robot behavior and mood-aware intervention strategies.

5.2.3 Thermal Sensing

To monitor user health and screen for potential fever symptoms, we integrated a thermal sensing module based on the FLIR Lepton 3.5 sensor, mounted on a PureThermal 3 USB board. This module is dedicated to non-contact skin temperature estimation, leveraging long-wave infrared (LWIR) imaging in a compact and low-power setup.

Hardware and Configuration

The thermal sensor operates with a resolution of 160x120 pixels and a field of view (FOV) of 57°, with radiometric accuracy sufficient for medical pre-screening in indoor environments. The PureThermal3 development board enables real-time USB video class (UVC) streaming of the thermal image, which is then processed using OpenCV and standard image processing libraries on the Jetson Xavier platform.

Detection Pipeline

Our implementation follows the methodology proposed by Deligiannidis et al. [31], where RGB-based face detection is used to localize the forehead region within the thermal frame. The maximum pixel value within this region is extracted and converted to Celsius using sensor calibration parameters. This value is considered the estimated body surface temperature.

The ROS node performs the following steps:

1. Acquires a synchronized RGB and thermal frame pair.
2. Detects the face and estimates the region of interest (ROI) corresponding to the forehead.
3. Extracts the maximum temperature within the ROI from the thermal image.
4. Publishes the result on a ROS topic `/thermal_temperature`.

Use in the System

Unlike other emotion-driven applications of thermal imaging, our system does not attempt to infer psychological states (e.g., stress or arousal) from thermal patterns. Instead, we adopt a purely physiological application of thermal sensing: estimating body temperature to detect high fever conditions.
The output of this module is fused with other signals (FER, SER, mmWave) in the mood and health estimation layer, described in a later section.

5.2.4 mmWave Radar Sensing

To complement visual and audio inputs with physiological measurements, we integrated a 60GHz mmWave radar module (Seeed Studio R60A) capable of detecting vital signs such as heartbeat, heart rate waveform, and breathing characteristics. This module enables contactless physiological monitoring even in low-light or partially occluded conditions.

Functionality and Configuration

The R60A radar communicates with the robot via a serial interface. The raw hexadecimal data stream encodes various physical metrics, including:

- Heartbeat detection (presence/absence)
- Heart rate (in beats per minute)
- Heart rate waveform (for visual analysis)
- Breathing patterns (normal, fast, slow, none)
- Body movement and target orientation
- Proximity status (within range / out of range)

ROS Integration

The mmWave node is implemented in Python and runs as a ROS node. It listens to the serial port, decodes the relevant packet headers (e.g., `535985020001` for heart rate), and publishes the extracted values to dedicated ROS topics:

- `/heartbeat` – `Int32`: raw heartbeat detection
- `/heartrate` – `Int32`: estimated heart rate in BPM
- `/heartratewaveform` – `Int32`: waveform component (e.g., for visualization)

These values are published in real-time and can be visualized in tools like `rqt_plot` or integrated with JSK RViz plugins for overlay monitoring.

Role in Multimodal Estimation

The mmWave radar provides physiological signals that are orthogonal to those extracted via vision or speech. It is especially useful for health monitoring in ambient scenarios where camera visibility is degraded or privacy is a concern. These measurements are fused with facial expression and speech emotion data to produce a robust estimation of the user's physical and emotional state.

5.3 Sensing Fusion

In multimodal perception, combining diverse sensory inputs is essential to improve the reliability of affective state estimation. In SUNNY, a custom ROS node was developed to fuse data from facial expression recognition (FER), speech emotion recognition (SER), thermal sensing, and mmWave-based vital signs (e.g., heart rate).

5.3.1 Fusion Strategy and Mood Estimation

The fusion process focuses primarily on aligning and merging FER and SER outputs, which are received in real-time via the custom message type `fac-`

`erec::People`. This message encapsulates all the per-user data extracted by the perception modules, including:

- Facial expression (FER) label
- Speech emotion (SER) label
- Body temperature (from thermal camera)
- Heart rate (from mmWave radar)
- Frame-based position and office coordinate
- Identity (if recognized)

The mood estimation algorithm then evaluates the consistency between FER and SER to compute an overall `mood` label and a corresponding `intensity` score in the range $[0.25, 1.0]$. The logic is rule-based and designed to capture:

- Strong agreement (same emotion $\Rightarrow$ intensity $= 1.0$)
- Partial agreement (one neutral, one emotional $\Rightarrow$ 0.5)
- Opposing emotions (e.g., happy vs. sad $\Rightarrow$ 0.25)
- Similar emotions (e.g., fear vs. sadness $\Rightarrow$ 0.75)

5.3.2 ROS Integration and Output Flow

The fusion node subscribes to the topic `/people_data_final` and runs continuously. For each detected person with valid data, it performs fusion and publishes the result to the interaction node via a real-time message.

This message is also saved in CSV format in two parallel databases:

1. A personal history file for each user
2. A daily log file saved under a folder named with the current date

5.3.3 Example Output and Application

An example of the CSV output is shown below:

```
Timestamp,ID,Name,Expression,FER,SER,Temperature(°C),Heartrate(bpm),
x,y,x1,y1,w,h,PositionX,PositionY,Mood,Intensity
2024/09/01 15:42:31,3,Fabio,Happy,Happy,Happy,36.8,75,231,112,...
```

This fused data is forwarded to the interaction node, which uses it to decide whether and how to engage the user. For example, if a user is detected as sad with high confidence, the robot may initiate a motivational message or suggest a break.

This fusion mechanism plays a critical role in linking the perception layer to the behavior layer, enabling real-time, context-aware social interaction.

5.3.4 Sensing Fusion

In multimodal perception, combining diverse sensory inputs is essential to improve the reliability of affective state estimation. In SUNNY, a custom ROS node was developed to fuse data from facial expression recognition (FER), speech emotion recognition (SER), thermal sensing, and mmWave-based vital signs (e.g., heart rate).

Fusion Strategy and Mood Estimation

The fusion process focuses primarily on aligning and merging FER and SER outputs, which are received in real-time via the custom message type `facerec::People`. This message encapsulates all the per-user data extracted by the perception modules, including:

- Facial expression (FER) label
- Speech emotion (SER) label
- Body temperature (from thermal camera)
- Heart rate (from mmWave radar)

- Frame-based position and office coordinate
- Identity (if recognized)

The mood estimation algorithm then evaluates the consistency between FER and SER to compute an overall `mood` label and a corresponding `intensity` score in the range $[0.25, 1.0]$. The logic is rule-based and designed to capture:

- Strong agreement (same emotion $\Rightarrow$ intensity = 1.0)
- Partial agreement (one neutral, one emotional $\Rightarrow$ 0.5)
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- Similar emotions (e.g., fear vs. sadness $\Rightarrow$ 0.75)

Related Work and Methodological Justification

Several prior works in affective computing, particularly in Human-Robot Interaction (HRI), have adopted rule-based or interpretable methods for multimodal emotion fusion. These include approaches that combine outputs from facial expression recognition (FER) and speech emotion recognition (SER) using decision-level logic, fuzzy systems, or confidence-based voting mechanisms.

Sahoo and Routray [32] proposed a rule-based decision fusion framework to integrate FER and SER, enhancing robustness in emotion recognition systems. Similarly, Alonso-Martín et al. [33] developed an HRI system that combines audiovisual signals using confidence-weighted decisions, demonstrating practical deployment in robots interacting with children. Khanna and Sasikumar [34] employed a MYCIN-style certainty factor method to combine emotion cues in a hybrid rule-based reasoning engine.

Compared to these approaches, our method is intentionally simple and interpretable: it encodes domain knowledge about emotional oppositions and similarities, and provides a continuous-valued intensity signal that reflects the agreement level between modalities. This aligns with early multimodal HRI studies, such as De Silva and Ng [35], who achieved promising results by combining FER and SER with hand-crafted fusion rules.

Future Work: While this rule-based approach is effective and transparent, it may be further improved. With more time and access to longitudinal interaction data, we could perform statistical tuning of thresholds or transition to a learned model that better captures the joint dynamics of FER and SER. Data-driven methods, such as neural fusion architectures or probabilistic graphical models, could help optimize emotional inference by learning from large-scale multimodal datasets. This remains a promising direction for future work.

Implementation Status (Prototype). The Circumplex projection (valence–arousal), the fever-threshold hysteresis, and the HR-driven arousal modulation described above were *not deployed* in the current SUNNY prototype at runtime. These components are part of the designed methodology and have been studied at a theoretical level (with preliminary offline tests), but were excluded from the on-robot software due to time constraints. Therefore, all experiments and results in this thesis rely on the discrete rule-based fusion (FER+SER) with the `mood`/`intensity` outputs described earlier. A full integration of the VA stream and health heuristics is planned as future work (see Capter 8).

5.4 Navigation and Spatial Awareness

The ability to autonomously navigate within office spaces is a core requirement for SUNNY, enabling the robot to reach interaction zones, avoid obstacles, and maintain natural positioning in human environments. This section describes the navigation pipeline, sensor configuration, and ROS integration employed in the system.

5.4.1 Hardware and Sensor Setup

SUNNY is equipped with two complementary LIDAR systems:

- **2D LiDAR (YDLIDAR TG30)**: Mounted at wheel height on the front of the robot, this sensor provides planar obstacle detection for real-time ground-level navigation. It publishes a LaserScan message on the `/scan_tg30` topic.
- **3D LiDAR (Livox MID-360)**: Mounted at the center of the robot's structure, this sensor enables detection of higher obstacles, such as desks, chairs, or people. It publishes 3D point clouds on the `/passthrough2/output` and `/passthrough4/output` topics.

These two data sources are fused in the obstacle layer of the local and global costmaps, enabling accurate obstacle avoidance in both the horizontal and vertical domains.

5.4.2 Navigation Stack and Move Base

SUNNY's navigation system is built around the standard `move_base` node in ROS Noetic. The navigation launch sequence is defined in a `launch` file, which initializes the full stack with all relevant parameters.

The core modules include:

- **Global Planner**: Implements long-range path planning using `global_planner/GlobalPlanner`.
- **Local Planner**: Executes real-time trajectory generation and obstacle avoidance via `teb_local_planner/TebLocalPlannerROS`, configured through `teb_local_planner_params.yaml`.
- **Costmaps**: Obstacle layers are configured in `local_costmap_params.yaml` and `global_costmap_params.yaml`, while shared settings (inflation radius, resolution, plugins) are centralized in `costmap_common_params.yaml`.
- **Robot Footprint and Kinematics**: The mecanum drive configuration and footprint polygon are defined in `mecanum3.xacro`. The footprint accounts for the robot's full contour to improve planning accuracy.

5.4.3 Mapping and Environment Setup

Although SUNNY does not perform real-time SLAM, a semi-autonomous mapping phase is executed at deployment. During this phase, an operator remotely drives the robot through the office environment using a joystick or keyboard interface. The 2D LiDAR data is used to generate a static occupancy grid map, which is then stored and loaded at runtime for localization and planning.

During normal operation, SUNNY navigates on this static map and performs real-time obstacle avoidance using both 2D and 3D LIDAR fused in the costmap obstacle layer.

5.4.4 ROS Integration

The navigation system integrates with the overall robot architecture via standard ROS topics and services. The most relevant topics include:

- `/move_base/goal`: Receives navigation goals issued by the interaction node.
- `/move_base/status`: Monitors goal execution status.
- `/scan_tg30`, `/passthrough2/output`, `/passthrough4/output`: Provide fused obstacle data.
- `/odom`, `/map`: Used for localization and path tracking.

These components ensure robust and responsive movement in a structured indoor environment, allowing SUNNY to interact autonomously with people throughout the office.

5.4.5 Limitations

The navigation system currently assumes a known, mapped environment. While real-time obstacle avoidance is supported through fused LIDAR data, global map changes (e.g., reconfiguration of furniture) require remapping.

Additionally, the robot does not yet support autonomous docking or recharging, which is identified as a potential future enhancement.

5.5 Interaction Management

One of the key objectives of SUNNY is to conduct natural and context-aware human-robot interactions (HRI) within office environments. To achieve this, we developed a dedicated ROS node, `conversation.py`, which coordinates perception modules, dialogue generation, movement, and interaction history handling.

Interaction Pipeline

The node subscribes to a high-level trigger event—typically derived from the detection of a person and mood-health estimation. Upon activation, the following steps are executed:

1. **Navigation to the user**: The node receives the estimated 2D position of the user in the office map (computed in earlier perception layers) and sends a goal to the `move_base` ROS action server.

2. **Visual Eligibility Check (CLIP)**: Upon arrival, an image from the robot's frontal camera is analyzed using a pre-trained CLIP (Contrastive Language–Image Pretraining) model to determine whether the person is facing the robot and appears visually available for interaction. Only if the visual check is successful the interaction proceeds.

3. **Conversation Setup**: The user's identity and preferred language (previously inferred through face recognition and filename parsing) are used to initiate a contextualized conversation with OpenAI's GPT model via API.

4. **Multilingual Prompting and TTS**: A language-specific prompt is constructed based on the user's status (e.g., mood, temperature, pre-

vious interactions). The GPT-generated response is synthesized using text-to-speech (TTS) and played back via the onboard speaker.

5. **User Response Handling**: After playback, the robot activates a microphone and records the user's spoken response. The audio file is transcribed using the Whisper ASR model and added to the conversation thread.

6. **Contextual Memory Logging**: The interaction is appended to a dedicated text log file, which stores the last three exchanges with that user. This log is then used to re-contextualize future interactions.

Personalization and Thread Management

A custom mapping script, `create_threads.py`, maintains a dictionary linking each user name to a unique conversation thread ID and preferred language. This enables language-aware and memory-persistent interactions using OpenAI's API, ensuring continuity across multiple encounters.

ROS Integration and Interoperability

The interaction node integrates seamlessly with other modules via standard ROS messaging:

- Subscribes to: `/people_data_final` (containing mood, health, position, and ID), `/face_expression`, `/ser_results`
- Publishes to: `/move_base/goal`, `/interaction_status`
- Records audio from: external microphone (DJI Mic or Anker, depending on setup)

This modular design allows the interaction node to operate autonomously while remaining synchronized with sensing, navigation, and logging components. The discussion will also be extended in Chapter 6

Limitations

The current interaction framework assumes an indoor, low-noise environment and requires face recognition and voice input for full personalization. Future improvements may include emotion-adaptive dialog strategies and long-term memory of interaction patterns.

5.6 Map & Marker Logging

Objectives

To make sensing and interaction interpretable in space and time, SUNNY overlays semantic information on the office map and logs per-user spatial events. This layer serves two goals: (i) *online* visualization in RViz (e.g., who is where, known vs. unknown persons), and (ii) *offline* analysis of spatial behavior (e.g., dwell, visit counts) aligned with mood/health estimates. In practice, the runtime RViz view renders identities and emotions over the map (Fig. 5.1), while daily logs are aggregated into spatial mood overlays for offline analysis (Fig. 5.2). The end-to-end pipeline is summarized in Fig. 5.3.

ROS Message Interfaces (Recorded in Experiments)

We leverage the following topics (see also Tables 5.1–5.2):

- `/known_marker`, `/unknown_marker` (`visualization_msgs/MarkerArray`): geometric/text markers for entities whose identity is *known* or *unknown*.
- `/name_marker` (`visualization_msgs/MarkerArray`): text labels with user names (and, optionally, attributes such as preferred language).
- `/people_positions` (`markers/Peop_pos`): custom message with 2D person positions in the `map` frame.
- `/people_data_final` (`facerec/People`): fused per-person record (FER, SER, temperature, HR, pose, mood, intensity).

- /map (nav_msgs/OccupancyGrid), /tf, /amcl_pose: map, transforms, and localization for consistent placement.

Online Marker Publisher (markers.cpp)

The markers.cpp node subscribes to /people_data_final and /people_-positions and publishes:

- **Identity/Status**: /known_marker for recognized users (e.g., name available from the FER/SFace pipeline) and /unknown_marker for unrecognized faces or anonymous tracks.
- **Name labels**: /name_marker as TEXT_VIEW_FACING markers positioned at the person's 2D pose (frame_id=map); see the live snapshot in Fig. 5.1.
- **Mood-aware styling**: color/scale can encode the fused mood and intensity (e.g., green for positive, blue neutral, red negative; scale factor $\propto$ intensity).

Markers use persistent id/ns to avoid flicker and to enable updates/removals. Time stamps and lifetime are set to keep the view responsive in RViz.

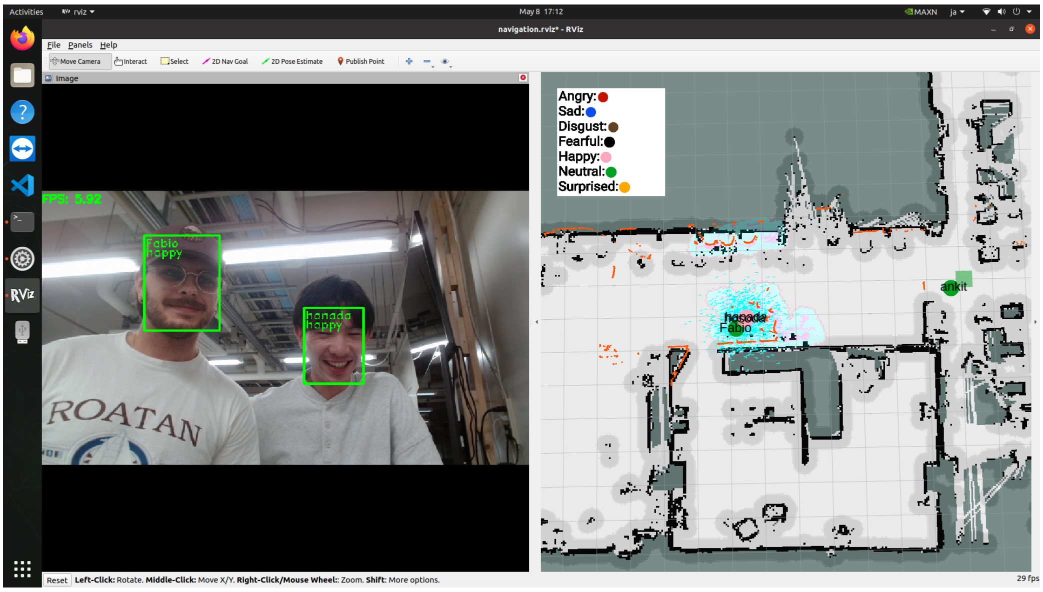

Figure 5.1: Runtime RViz view with identity and emotion markers over the static map; the legend indicates the color mapping per emotion.

Daily Marker Logger (`daily_markers.cpp`)

The `daily_markers.cpp` node mirrors the online view into structured files for longitudinal analysis. For each day (`YYYY-MM-DD`), it creates a folder and appends per-user entries (timestamp, identity, pose, mood, intensity), aligned with the fused stream:

- **Per-user logs**: individual CSVs under a user-centric directory (e.g., `.../People/Name.csv`).
- **Per-day logs**: daily CSVs under `.../People_date/YYYY-MM-DD/`.

This matches the logging strategy used by the fusion node (Section 5.3) and enables reproducible offline analytics feeding the spatial overlays in Fig. 5.2.

Offline Map Analysis (`map_analysis.py`)

The `map_analysis.py` utility ingests daily/user CSV logs and the occupancy grid to compute and export:

- **Visit/dwell metrics**: per user and per area (e.g., counts, cumulative dwell time).
- **Mood overlays**: aggregation of mood/intensity by location (heatmaps or discrete bins); an example is shown in Fig. 5.2.
- **Exports**: summary CSV and optional PNG overlays aligned to the map resolution/origin (uses `map.yaml` for metadata).

The script can be parameterized with input folders and produces analysis artifacts referenced in the Experiments/Results chapters.

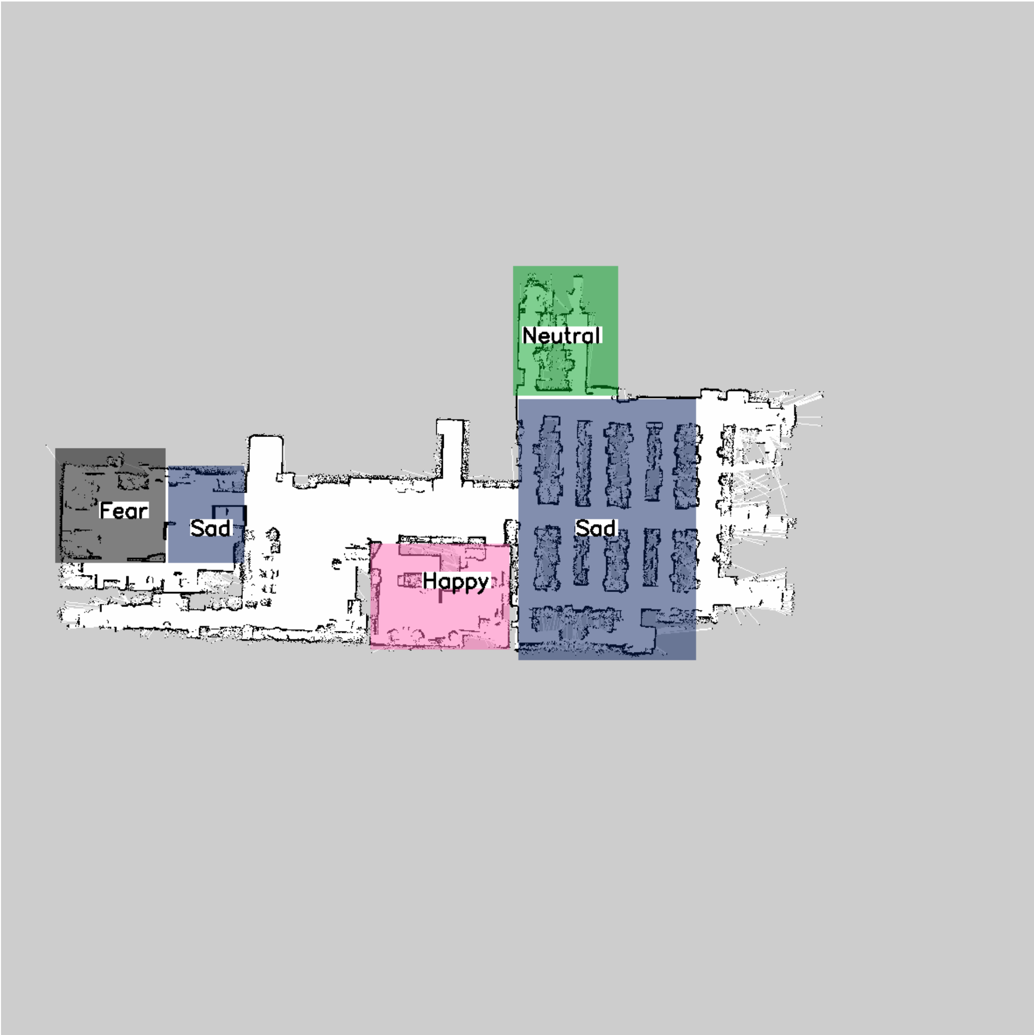

Figure 5.2: Offline daily mood overlay: each area is colored by the dominant mood aggregated from the daily's fused observations.

Marker Semantics and Styling

We adopt the following conventions for clarity in RViz:

- **Namespaces**: `ns = known`, `unknown`, `name`.
- **IDs**: stable incremental IDs per namespace; user-specific IDs reused across updates.

- **Color map (mood)**: `Happy/Positive` → green, `Neutral` → blue, `Negative (e.g., Sad/Angry)` → red; alpha = 0.8.
- **Scale**: proportional to intensity `[0.25, 1.0]` (Section 5.3).

ROS Flow (Markers and Analysis)

Figure 5.3 shows the end-to-end data flow for spatial logging and visualization, linking the fused stream to the online RViz markers (Fig. 5.1) and to the offline daily overlays (Fig. 5.2).

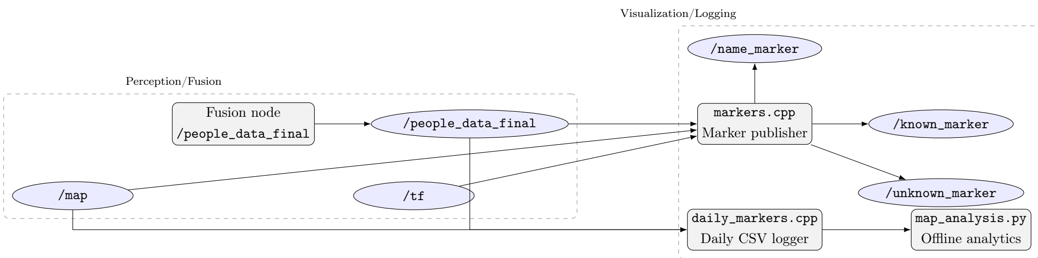

Figure 5.3: Marker/Map pipeline. The fusion output drives online RViz markers and daily logs; offline analysis aggregates spatial and mood metrics on the occupancy grid.

5.7 ROS Architecture

This section describes how SUNNY's software modules interoperate through ROS topics and actions during runtime. The integration follows a layered, message-centric design: perception nodes publish sensor-derived observations; a fusion node aggregates multimodal cues into high-level user state; the interaction node decides whether and how to engage; the navigation stack (AMCL + move_base with TEB) executes motion goals while maintaining safe trajectories on the office map.

5.7.1 Node Groups and Data Flow

Perception (Sensors → Features). RGB FER/SER and thermal modules publish visual frames and affective/physiological estimates; the mmWave node publishes vital signs. These streams are combined into a structured record per person:

- Images & thermal: `/img_face` (`sensor_msgs/Image`), `/thermal_img` (`sensor_msgs/Image`), `/people_temp` (`thermal/ThermalArray`).
- Emotion/vitals: `/ser_results` (`std_msgs/String`), `/heartrate` (`std_msgs/Int32`), `/heartbeat` (`std_msgs/Int32`).
- Multimodal bundle: `/people_data` (`facerec/People`).

Fusion (Features → Mood/Health). The fusion node subscribes to `/people_data`, applies the rule-based logic (agreement/similarity/opposition between FER and SER with intensity scaling), attaches temperature and HR, and publishes:

- `/people_data_final` (`facerec/People`) — the per-person record with *mood* and *intensity*.

Interaction (State → Dialogue). The conversation manager subscribes to `/people_data_final` and, when appropriate, triggers a CLIP-based availability check on the current frame. If eligible, it logs timestamps (`/start_conversation_timestamps`, `/conversation_timestamps`, `/end_conversation_timestamps`) and runs the multilingual dialogue pipeline (GPT/TTS/ASR). Interaction history is kept per user.

Navigation (Goals → Motion). High-level behaviors send goals to `/move_base/goal`. AMCL provides localization on the static map; the global and local planners (GlobalPlanner + TEB) operate over fused obstacle layers from 2D/3D LIDAR. Execution and feedback flow through standard move_base action topics.

Mapping/Localization and Visualization. The system consumes /map, /odom, /tf, and visualizes trajectories and pose arrays (e.g., TEB markers). Marker arrays (/known_marker, /unknown_marker, /name_marker) overlay semantic information on the map for analysis.

5.7.2 Key Topics and Message Types

Topic	**Type**
/people_data	facerec/People
/people_data_final	facerec/People
/ser_results	std_msgs/String
/heartrate, /heartbeat	std_msgs/Int32
/img_face	sensor_msgs/Image
/thermal_img	sensor_msgs/Image
/people_temp	thermal/ThermalArray
/scan_tg30, /scan	sensor_msgs/LaserScan
/map, /map_metadata	nav_msgs/OccupancyGrid/MapMetaData
/odom	nav_msgs/Odometry
/tf, /tf_static	tf2_msgs/TFMessage
/move_base/goal	move_base_msgs/MoveBaseActionGoal
/move_base/status	actionlib_msgs/GoalStatusArray
/move_base/...plan, costmap	nav_msgs/Path/OccupancyGrid
/known_marker, /unknown_marker	visualization_msgs/MarkerArray
/name_marker, /people_positions	visualization_msgs/MarkerArray
/start_conversation_timestamps	std_msgs/Time
/conversation_timestamps	std_msgs/Time
/end_conversation_timestamps	std_msgs/Time

Table 5.1: ROS topics and their message types recorded during the experiments.

Topic	Role
/people_data	Multimodal per-person bundle (FER, SER, temperature, heart rate, position).
/people_data_final	Fused record with mood and intensity for downstream interaction.
/ser_results	SER categorical emotion (e.g., Angry, Sad, Happy, Surprise, Fear, Disgust, Contempt, Neutral).
/heartrate, /heartbeat	Heart rate (bpm) and beat presence from mmWave sensing.
/img_face	RGB face stream for FER/recognition context.
/thermal_img	Thermal image for fever screening.
/people_temp	Per-person thermal readings aggregated from the thermal sensor.
/scan_tg30, /scan	2D LiDAR scans used for ground-level obstacle detection.
/map, /map_metadata	Static occupancy grid and metadata for localization and planning.
/odom	Robot odometry for local planning and tracking.
/tf, /tf_static	Frame transforms for all modules (mapping, navigation, perception).
/move_base/goal	Navigation goals issued by high-level behaviors (interaction).
/move_base/status	Status feedback from the move_base action server.
/move_base/... plan, costmap	Global/local plans and costmaps used by the navigation stack.
/known_marker, /unknown_marker	Semantic markers rendered in RViz for spatial context.
/name_marker, /people_positions	Overlay of user names and positions on the map.
/start_conversation_timestamps	Interaction timing (start).
/conversation_timestamps	Interaction timing (events).
/end_conversation_timestamps	Interaction timing (end).

Table 5.2: ROS topics and their functional roles in SUNNY's runtime architecture.

5.7.3 Node Graph

We derived the node graph *offline* from the recorded rosbag. Specifically, we parsed the bag's topic/type index and the `/rosout` / `/rosout_agg` logs with a custom Python script to reconstruct the set of active nodes and the topics used during the experiment. The resulting DOT description (grouped into functional clusters: *Perception*, *Fusion*, *Interaction*, *Navigation*, *Visualization*) was then rendered with Graphviz to a PNG and included as Fig. 5.4 (`figures/ros_node_graph.png`).

This offline graph faithfully reflects the dataflow observed in the recording (topics and their message types/counts), while manual highlighting emphasizes key pipelines (e.g., `Fusion` → `/people_data_final` → `conversation.py` → `/move_base/goal`). Note that, unlike a live `rqt_graph`, publisher/subscriber directionality cannot always be inferred from logs alone; therefore, arrows in the diagram denote the intended flow of information across modules rather than the transient transport connections present at runtime.

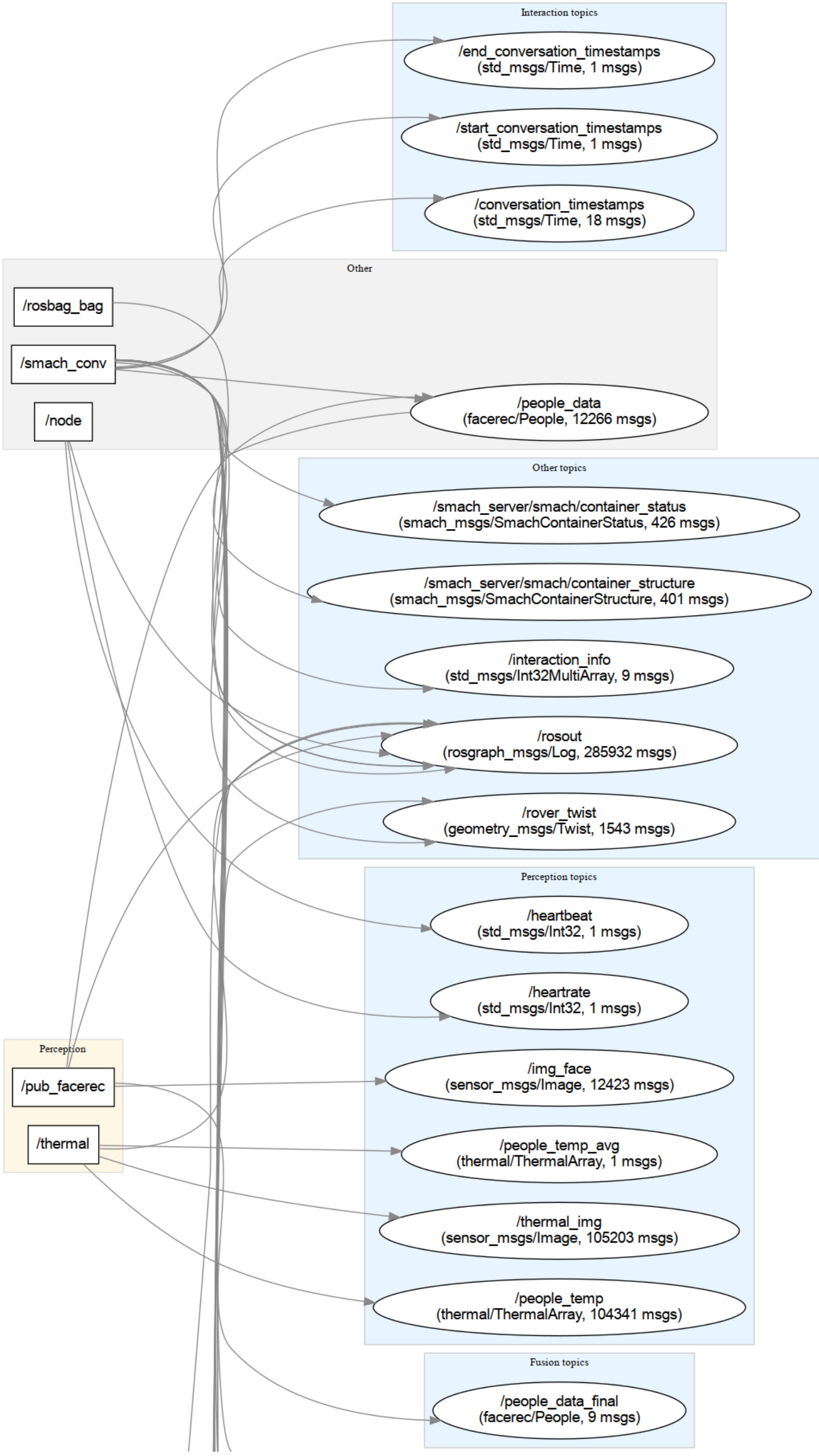

Figure 5.4: ROS node graph of SUNNY derived from rosbag topics.

Chapter 6

Intervention

6.1 Goals and Design Principles

The intervention layer translates fused perception outputs (cf. Section 5.3) into short, context-aware actions that can support users' well-being in office environments. SUNNY is designed to be:

- **Helpful but lightweight**: brief micro-interventions, no clinical claims.
- **Non-invasive and respectful**: engage only when appropriate; avoid over-interaction (per-user cool-down).
- **Privacy-preserving**: log only fused summaries; personal user-robot interaction; no log shared among the different users.

6.2 Runtime Modes (for Experiments)

We implement two runtime modes that correspond to the experimental conditions:

With Data Feedback The LLM receives structured context (name, FER, temperature) and adapts its replies accordingly (reinforce positive affect, suggest breaks for negative, warn if temperature $> 38\,^{\circ}\mathrm{C}$).

No Data The LLM responds only to the user message, keeping health tone generic.

These modes are operationalized via two prompt templates (Section 6.4) and are compared in the Experiments chapter.

6.3 Personalization

When available, `name` and `preferred language` (from the FER/SFace pipeline) tailor greetings and phrasing, also retrieving information from a log of the last 3 interactions had with the same user; otherwise, the system falls back to neutral English and eventually changes language, answering with the last language spoken by the user.

6.4 Prompt Templates (LLM Policies)

The conversation node (`conversation.py`) drives the LLM with template policies. We report the exact instructions used in the experiments.

Office Robot (no data)

Listing 6.1: Office Robot (no data) policy

```
-You have a role in maintaining and promoting employee health in the
    ↪ office.
-Please engage in natural interaction with the user’s input.
-Always send as first message "Hi" + user’s name.
-Please choose an appropriate topic from previous conversations or talk
    ↪ about a new topic yourself.
-Please use the name of the person you are talking to, during interactions
    ↪  in a natural way.
-Always change language based on the user’s one.
-Do not add extra polite language or phrases prompting for follow-ups and
    ↪ do not send tedious and long messages.
-If the person has negative emotion, tell them to take a break and free up
    ↪  their mind (talking with some office colleagues in the cafeteria
    ↪ can be an idea).
-If a person’s emotions are ’neutral’, do not mention that he or she is ’
    ↪ neutral’.
-if person’s temperature is above 38 advise them about that, and that they
    ↪  should check if they have high fever or not with a thermometer.
-If in a message the speaker says "stop the communication", "see you" and
    ↪ "bye" (or anything with the same meaning) return a " False" as a
```

```
    ↪ response.

temperature 0.80, top p 1.00.
```

Office Robot (with Data feedback)

Listing 6.2: Office Robot (with Data feedback) policy

```
The messages we are sending you are in the following format:
"[name, FER, temperature]
user message"
Use this information to tailor your responses. For example, address the
    ↪ user by name and adjust your interaction based on their FER and
    ↪ temperature.

-Your goal is to let people change FER. You have to positively impact the
    ↪ mood of people. don't stop the conversation until they're happy and
    ↪  try to arouse a positive emotion while interacting. Please give
    ↪ priority to this and try to maintain the attention of the person
    ↪ you are talking with. Try to be funny, do jokes, and try to ask
    ↪ about themselves. Your goal is to get know these people and help
    ↪ them during the day.

- If anybody ask about you, your name is  SUNNY and you have a role in
    ↪ maintaining and promoting employee health and mood in the office.
- Always Introduce yourself if no previews conversation is uploaded.
-Please engage in natural interaction with the user's input.
-Always send as first message "Hi" + user's name.
- you can reference previous conversations to provide continuity, but
    ↪ avoid repetitive patterns. Vary your phrasing and approach in each
    ↪ conversation, even if you're reinforcing the same concepts or goals
    ↪ . Use creative language to keep interactions fresh and personalized
    ↪ .
-Please use the name of the person you are talking to, during interactions
    ↪  in a natural way.
-Always change language based on the user's one.
-Do not add extra polite language or phrases prompting for follow-ups and
    ↪ do not send tedious and long messages. Prefer a cross-talk style of
```

```
    ↪ communication instead of sending long messages.
-Don't ever use emoji.
-if person's temperature is above 38 advise them about that, and that they
    ↪ should check if they have high fever or not with a thermometer.
-If in a message the speaker says "stop the communication", "see you" and
    ↪ "bye" (or anything with the same meaning), respond with just "
    ↪ False".

mood:
-if the person has positive emotion, reinforce positivity, make jokes,
    ↪ try to be smart and funny.
-If the person has negative emotion, suggest helpful actions. For instance
    ↪ , if they are stressed or upset, encourage them to take a break and
    ↪ suggest light conversation with colleagues in the cafeteria. If
    ↪ they seem overwhelmed, recommend a short break to clear their mind.
-If a person's emotions is 'neutral', do not mention that he or she is '
    ↪ neutral'. Instead, engage naturally and try to make them smile.

temperature 1.20, top p 1.00, gpt-4o
```

6.5 Ethics, Privacy, and Safety

Scope and disclaimers. SUNNY provides well-being *prompts*, not medical advice. Health-related outputs are generic (rest/check temperature), never diagnostic.

Data minimization. Only fused summaries are logged (mood/intensity, optional VA, smoothed temperature/flag, HR, 2D pose). No raw audio/RGB/thermal is stored.

Privacy-by-design. Recognition uses feature comparison on a consented set; no runtime storage of raw face images.

6.6 Implementation Status & Link to Experiments

The policies above are implemented through `conversation.py` and the two prompt templates. Advanced options (valence–arousal projection, HR-driven arousal modulation, fever hysteresis) were designed and tested offline but *not* deployed on-robot and are therefore evaluated only qualitatively. The Experiments chapter will compare the two modes (*with data feedback* vs. *no data*) on interaction quality and user-reported outcomes.

CHAPTER 7

Experiments and Results

7.1 Objectives and Hypotheses

The experimental phase of this thesis was designed to directly address the research questions defined in Chapter 3. In particular, the first set of experiments evaluates the reliability of SUNNY's multimodal, non-invasive mood and health estimation pipeline, while the second set examines the effectiveness of robot-led interventions in shaping user affect and raising health awareness in office-like scenarios.

From these research questions, the following working hypotheses were derived:

- **H1 (Mood Estimation).** The fused FER+SER module, complemented by thermal and mmWave cues, can reliably approximate human judgments of affective state in real time.
- **H2 (Intervention Effect).** Interactions with SUNNY result in a measurable improvement in users' reported mood immediately after the session.
- **H3 (Data Feedback Benefit).** Providing the LLM with fused sensing context (name, FER, temperature) increases the impact and engagement of interventions compared to the context-free mode.

7.2 Mood Estimation Experiment

Experimental Setup

Participants were seated one at a time at a fixed point in the office environment. The robot approached and interacted briefly, while each participant was instructed to exhibit a specific mood (happy, sad, angry, neutral, etc.). The SUNNY robot's multimodal sensing modules (FER, SER, thermal, mmWave) ran in real time, generating fused records (`/people_data_final`) as described in Section 5.3.

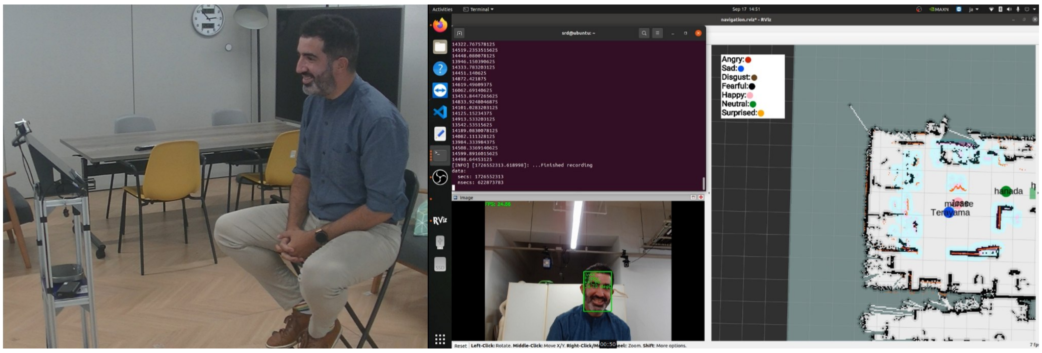

Figure 7.1: Example of the Mood Estimation Experiment interaction.

Data Collection

The robot stored for each frame:

- FER and SER categorical labels,
- fused `mood` and `intensity`,
- temperature estimate and HR value (when available),
- user ID and position.

Additionally, a survey was administered to an independent group of observers (not participants) who watched the recorded videos. For each clip, observers annotated the mood they perceived the user to be displaying. This survey acts as a “ground truth by perception” against which the robot’s estimation can be compared.

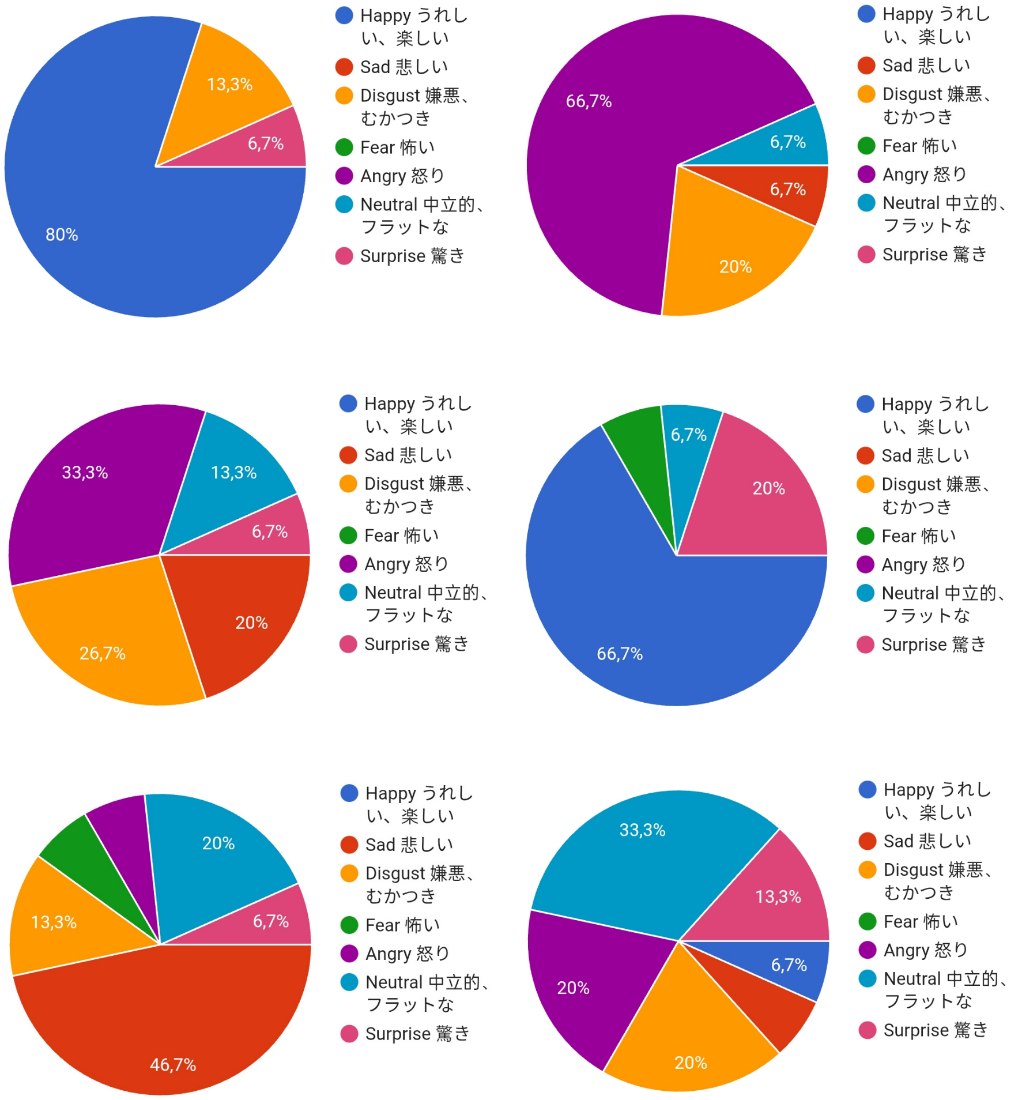

Figure 7.2: Example of the first form answers for six of the recorded interactions

Data Analysis

Robot mood estimations were aligned interaction-by-interaction with observer votes. Accuracy, precision, recall, and Cohen's κ were computed to evaluate agreement. Intensity scores were compared against observers' confidence ratings when available. Statistical analysis was conducted using paired comparisons between robot-estimated moods and human-labeled moods.

Results

Evaluation metrics. We evaluate SUNNY's predictions against survey labels using the metrics below.

- **Accuracy** — Overall fraction of correct predictions; single-number summary of performance (in single-label multi-class it equals micro-F1).
- **Precision**$_c$ — For class c: proportion of predicted-c that are truly c; penalizes false positives for class c.
- **Recall**$_c$ — For class c: proportion of true-c that are predicted c; penalizes false negatives for class c.
- **F1**$_c$ — Harmonic mean of Precision$_c$ and Recall$_c$; balances precision/recall for class c.
- **Macro-average** — Unweighted mean over classes; mitigates class imbalance by giving each class equal weight.
- **Cohen's** κ — Chance-corrected agreement; quantifies agreement beyond chance between robot and survey labels.
- **95% CI (Wilson)** — Uncertainty on accuracy as a binomial proportion; more reliable than Wald for small N or extreme $\hat{p}$.

Let K be the number of classes and $M \in \mathbb{N}^{K\times K}$ the confusion matrix with rows = ground truth (survey) and columns = robot predictions. With $N = \sum_{i,j} M_{ij}$ and M_{ii} the diagonal terms:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{K} M_{ii}.$$

For a class c:

$$\begin{aligned}
\text{Precision}_c &= \frac{M_{cc}}{\sum_{i=1}^{K} M_{ic}}; \\
\text{Recall}_c &= \frac{M_{cc}}{\sum_{j=1}^{K} M_{cj}}; \\
F1_c &= \frac{2\,\text{Precision}_c\,\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.
\end{aligned}$$

We report *macro*-averages as the mean across classes (useful with class imbalance). In single-label multi-class tasks, *micro*-F1 coincides with accuracy.

To assess agreement beyond chance we use Cohen's κ:

$$\begin{aligned}
p_o &= \frac{1}{N}\sum_{i} M_{ii}; \\
p_e &= \frac{1}{N^2}\sum_{i}\Big(\sum_{j} M_{ij}\Big)\Big(\sum_{j} M_{ji}\Big); \\
\kappa &= \frac{p_o - p_e}{1 - p_e}.
\end{aligned}$$

Where Cohen's κ is a *chance-corrected* measure of agreement between two labelers ("raters") on categorical data [36]. Unlike plain accuracy—which is just the fraction of matches—κ discounts the portion of agreement that would be expected *by chance* given each rater's marginal label distribution. Hence, κ answers: *how much better than chance* the observed agreement is.

In our multi-class setting, the same definition applies using:

$$p_o = (\text{observed agreement}),$$

$$p_e = (\text{chance agreement}),$$

Here, p_e is computed from the product of the two raters' marginal probabilities per class. Interpretation: $\kappa = 1$ means perfect agreement; $\kappa = 0$ means agreement is no better than chance; $\kappa < 0$ indicates *systematic* disagreement (worse than chance).

Interpretation. Thresholds are heuristic but commonly reported as in Landis & Koch [37]:

κ range	Qualitative strength of agreement
< 0	Poor
0.00–0.20	Slight
0.21–0.40	Fair
0.41–0.60	Moderate
0.61–0.80	Substantial
0.81–1.00	Almost perfect

Table 7.1: Commonly used (heuristic) interpretation of Cohen's κ [37].

95% confidence interval for accuracy (Wilson). For an observed accuracy p_o over N trials, we use the Wilson score interval (better coverage than Wald for small N/extreme p_o [38, 39]:

$$\mathrm{CI}_{\mathrm{Wilson}} = \frac{p_o + \frac{z^2}{2N} \ \pm \ z\sqrt{\frac{p_o(1-p_o)}{N} + \frac{z^2}{4N^2}}}{1 + \frac{z^2}{N}}.$$

Here z is the *critical value* of the standard normal distribution corresponding to the desired confidence level. It is obtained as the $(1-\alpha/2)$ quantile of the $\mathcal{N}(0,1)$ distribution, where α is the complement of the confidence level. For a 95% interval, $\alpha = 0.05$ and thus

$$z = \Phi^{-1}(1 - \tfrac{\alpha}{2}) = \Phi^{-1}(0.975) \approx 1.96,$$

with Φ^{-1} denoting the inverse cumulative distribution function of the standard normal. For a 99% CI, $\alpha = 0.01$ and $z \approx 2.576$; for 90%, $z \approx 1.645$.

Main-mood (6 classes) results. In our case $N = 34$ labeled segments, overall accuracy is **58.82%**:

$$p_o = \frac{20}{34} \approx 0.5882 \quad (58.82\%).$$

Leading to:

$$\text{Wilson 95\% CI} : [42.2, 73.6]$$

with $\kappa = 0.42$ (moderate agreement)(cf. Table 7.1). Macro-averaged results show **Precision = 0.693**, **Recall = 0.342**, and **F1 = 0.299**, reflecting the limited coverage of rarely predicted classes. Per-class metrics (with support in parentheses) are as follows:

- `Happy` (11): Recall = 100.0%, Precision = 57.9%, F1 = 73.3%.
- `Neutral` (10): Precision = 100.0%, Recall = 30.0%, F1 = 46.2%.
- `Sad` (8): Precision = 50.0%, Recall = 75.0%, F1 = 60.0%.
- `Angry` (3), `Fear` (1), `Surprise` (1): Recall = 0 (no predicted instances), consistent with low support and known confusions among negative emotions.

These results highlight strong detection of `Happy` and `Sad`, while performance drops significantly for minority classes.

	Predicted by Robot					
Survey (True)	`Angry`	`Fear`	`Happy`	`Neutral`	`Sad`	`Surprise`
`Angry`	0	0	0	0	3	0
`Fear`	0	0	0	0	1	0
`Happy`	0	0	11	0	0	0
`Neutral`	0	0	5	3	2	0
`Sad`	0	0	2	0	6	0
`Surprise`	0	0	1	0	0	0

Table 7.2: Confusion matrix (Main Mood classes). Rows = survey (true), columns = robot.

Class	Precision	Recall	F1	Support
`Angry`	—	0.0%	0.0%	3
`Fear`	—	0.0%	0.0%	1
`Happy`	57.9%	100.0%	73.3%	11
`Neutral`	100.0%	30.0%	46.2%	10
`Sad`	50.0%	75.0%	60.0%	8
`Surprise`	—	0.0%	0.0%	1

Table 7.3: Per-class metrics (Main Mood). "—" indicates undefined precision (no predicted instances).

Positive/Neutral/Negative (PNN) results. Grouping emotions into {Positive, Neutral, Negative} yields higher robustness: accuracy **73.53%** (95% CI: [56.9, 85.4]) and $\kappa = 0.594$ (moderate-to-substantial agreement). Macro-averaged metrics are **Precision = 0.812**, **Recall = 0.711**, and **F1 = 0.687**, indicating more balanced performance than in the six-class case. Per-class metrics (with support in parentheses) are as follows:

- `Positive` (12): Recall = 100.0%, Precision = 66.7%, F1 = 80.0%.
- `Neutral` (10): Precision = 100.0%, Recall = 30.0%, F1 = 46.2%.
- `Negative` (12): Precision = 76.9%, Recall = 83.3%, F1 = 80.0%.

These results show robust separation between `Positive` and `Negative` moods, while `Neutral` remains the most challenging class, often confused with neighboring categories.

	Predicted by Robot		
Survey (True)	`Positive`	`Neutral`	`Negative`
`Positive`	12	0	0
`Neutral`	4	3	3
`Negative`	2	0	10

Table 7.4: Confusion matrix (Positive/Neutral/Negative).

Class	Precision	Recall	F1	Support
`Positive`	66.7%	100.0%	80.0%	12
`Neutral`	100.0%	30.0%	46.2%	10
`Negative`	76.9%	83.3%	80.0%	12

Table 7.5: Per-class metrics (PNN).

Summary. SUNNY reliably separates *positive* vs. *negative* moods (PNN accuracy ≈74%, $\kappa \approx 0.59$), with lingering ambiguity around *neutral* displays—consistent with known annotator uncertainty for neutral affect in brief, office-like interactions. In the six-class setting, strong performance on `Happy` and `Sad` contrasts with limited coverage of `Angry`/`Fear`/`Surprise`, largely due to low support and the system's conservative decision policy.

In total, **36 recorded interactions** were used for evaluation, and judgments were collected from a group of a little more than **20 external observers**. A larger pool of interactions and observers would have enabled finer tuning of the model's parameters and of the mood detection module, potentially leading to more robust and generalizable experimental results. These findings therefore support the use of a PNN abstraction during online interaction, while finer-grained labels may require larger datasets or learned fusion calibrated on longitudinal logs.

7.3 Intervention Experiment

Experimental Setup

Participants were recruited from the office/lab environment and interacted with SUNNY under two different LLM-based policies described in Chapter 6:

1. **No Data**: LLM received only the user message, ignoring FER/temperature context.

2. **With Data Feedback**: LLM received structured context (name, FER, temperature).

To control for order effects, conditions were administered in a counterbalanced sequence (AB/BA), so that half of the participants started with condition (1) followed by (2), while the other half followed the inverse order. Unlike fixed-length trials, the duration of each interaction was not predetermined. Sessions naturally ended when the participant decided to terminate the conversation by explicitly saying goodbye to the robot. A washout period of at least one full day separated the two conditions for every participant. This ensured that potential residual effects of the first interaction (e.g., mood shifts induced by the robot) did not carry over into the second session, providing a cleaner comparison between the two modes of interaction.

Data Collection

Administration protocol. After each interaction in both experimental conditions (*no data feedback* and *with data feedback*), participants completed the same post-interaction questionnaire. The form included a *Name* field to match responses across days and link them to robot logs (duration, turn count, intervention policy and timestamps), used solely for research analysis.

Instrument structure (0–10 scales unless noted). The questionnaire comprised the following item groups:

- **Affect states (categorical).** Self-reported mood at *start*, *during*, and *end* of the interaction: {`Happy, Sad, Angry, Neutral, Surprised, Afraid, Disgusted`}.
- **Affect dynamics.** Perceived mood change (0–10); perceived influence of the robot's behavior on mood (0–10); emotional engagement during the interaction (0–10).
- **Perceived adaptivity/understanding.** Robot understood my emotional state (0–10); robot adapted its behavior to my emotion (0–10).
- **Interaction quality and acceptance.** Overall interaction rating (0–10, anchor: 5 "adequate"); comfort working with/alongside a robot

(0–10, anchor: 5 “adequate”); willingness to work in a robot-enabled environment (0–10); perceived reliability of the robot (0–10); clarity of robot movements (0–10, anchor: 5 “adequate”).

- **Perceived usefulness and usability.** Helpfulness for my job (0–10); contribution to improving office environment (0–10); ease of use (0–10); satisfaction with the robot meeting expectations (0–10).

- **Environmental robustness expectation.** Expected change in robot behavior in unstructured environments (0–10; anchor: 5 “no change”, > 5 improved, < 5 worsened).

- **Open-ended feedback.** Description of mood change; suggestions about what the robot should have done differently; overall impression (free text).

Robot-side logging and alignment. Concurrently, the robot logged for each session: conversation duration, number of user turns, and intervention policy in use; logs were timestamped and linked to questionnaire responses via participant name and session time. This enables joint analyses (e.g., whether longer or more interactive sessions correlate with engagement or mood change).

Scoring and preprocessing. To prepare data for within-subject comparisons between conditions:

- **Categorical moods.** We analyze both the original 6-class labels and a grouped *Positive/Neutral/Negative (PNN)* abstraction. For intervention effects, we compare *end* vs. *start* self-reported mood per session, and between conditions.

- **Likert-type items (0–10).** When needed, scores are normalized to $[0, 1]$ for effect-size comparability. Items with anchors at 5 (“adequate”) are interpreted with 5 as a neutral baseline.

- **Signed-change items.** For questions with anchored neutrality at 5 (e.g., "improved/worsened"), we compute a centered score $s^\star = s - 5$ so that positive values indicate improvement.
- **Within-subject contrasts.** For each participant, we compute $\Delta =$ with-data − no-data on matched items (e.g., overall rating, engagement, perceived influence, etc.).
- **Statistical tests.** Primary analyses use paired t-tests on Δ; if normality is violated, we report Wilcoxon signed-rank. We include effect sizes (d_z for paired designs) and 95% CIs for mean differences.

Rationale and mapping to outcomes. The categorical mood triplet (start/during/end) operationalizes perceived affect trajectory; the adaptivity/understanding items probe whether the policy variant (with vs. without sensing feedback) is *perceived* as responsive; acceptance/usefulness/usability quantify user-centered outcomes relevant for deployment in office settings; robot-side logs allow exploratory correlations between behavior and outcomes.

Data Analysis

We adopt a within-subject analysis contrasting the two policies (*with data feedback* vs. *no data feedback*). For each participant and each matched item, we compute the paired difference $\Delta =$ with-data−no-data. Prior to inference, we inspect data quality (missing, outliers) and distributional assumptions on Δ (Shapiro–Wilk). If normality holds, we apply a paired t-test; otherwise, we report a Wilcoxon signed-rank test. We provide effect sizes (d_z for paired t; r for Wilcoxon) and two-sided 95% confidence intervals for mean differences (when t-tests are applicable). Primary outcomes include: overall interaction rating, emotional engagement, perceived influence on mood, perceived understanding/adaptivity, as well as the signed affect change in valence (Δv) derived from self-reported start/end moods. Robot-side logs (duration, user turns) are analyzed analogously to explore behavioral correlates. Unless otherwise noted, $\alpha = 0.05$ (two-tailed). We also check for possible order effects (AB vs. BA) descriptively; the main contrasts remain within-subject.

Results

Evaluation metrics. We analyze a within-subject contrast between the two policies (*with data feedback* vs. *no data feedback*) using paired statistics on per-participant differences. For each outcome and for each participant $i = 1, \ldots, n$ (where n is the paired sample size for that outcome), we define:

$$\begin{aligned} d_i &= \text{with-data}_i - \text{no-data}_i, \\ \bar{d} &= \frac{1}{n}\sum_{i=1}^{n} d_i, \\ s_d &= \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2}. \end{aligned}$$

Where: d_i is the *paired difference* for participant i (score under the *with-data* condition minus score under the *no-data* condition); $\bar{d}$ is the *sample mean* of the differences; s_d is the *sample standard deviation* of the differences.

- **Assumption check (normality).** We assess the normality of $\{d_i\}$ with the *Shapiro–Wilk* test, whose null hypothesis is that the paired differences come from a normal distribution. The test statistic is

  $$W = \frac{\left(\sum_{i=1}^{n} a_i\, d_{(i)}\right)^2}{\sum_{i=1}^{n}(d_i - \bar{d})^2},$$

  where $d_{(i)}$ are the ordered differences (from smallest to largest), $\bar{d}$ is the sample mean of the differences, and the weights a_i are constants (depending on n) derived from the expected values and covariance matrix of normal order statistics. Values of W close to 1 indicate data consistent with normality; smaller W suggests departures from normality. A *p-value* is obtained from W under the null.

  Decision rule and definitions:

- *p-value:* the probability, assuming the null hypothesis (normality) is true, of observing a test statistic at least as extreme as the one computed from the sample. A small p-value indicates evidence against the null.
- *Significance level α:* the pre-chosen Type I error threshold (probability of rejecting a true null). Here we use $\alpha = 0.05$ (5%).

If the p-value satisfies $p \geq \alpha$ (i.e., we do not reject normality), we proceed with a paired t-test; otherwise, we use the Wilcoxon signed-rank test. (Note: failing to reject normality does not prove normality; it only indicates insufficient evidence to deem the departures from normality statistically significant at level α.)

- **Paired t-test (mean difference).** When normality is plausible, the test statistic is:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad \mathrm{df} = n - 1,$$

testing the null hypothesis $H_0 : \bar{d} = 0$ (two-tailed). *Where:* t is the Student t-statistic; df are degrees of freedom.

  - *Effect size (paired):*

$$d_z = \frac{\bar{d}}{s_d},$$

  i.e., the standardized mean difference for paired designs (*Cohen's* d using the SD of differences). Heuristics: small ≈ 0.20, medium ≈ 0.50, large ≈ 0.80.

  - *95% CI for the mean difference:*

$$\bar{d} \pm t_{n-1,\,1-\alpha/2} \frac{s_d}{\sqrt{n}},$$

  where $t_{n-1,\,1-\alpha/2}$ is the critical value of the t-distribution with $n-1$ df at two-tailed level α (here 0.05). This interval indicates the range in which the true mean difference lies with 95% confidence.

- **Wilcoxon signed-rank test (median shift).** If normality is doubtful, we compare the median of differences via signed ranks: we rank

$|d_i|$ (ignoring zeros), apply the signs of d_i, and sum the signed ranks to obtain the statistic W. Two-sided p-values are computed (exact or via normal approximation).

– *Effect size (rank-based):*

$$r = \frac{Z}{\sqrt{n}},$$

where Z is the standard normal deviate corresponding to W under the normal approximation, and n is the number of paired observations included. Heuristics: small ≈ 0.10, medium ≈ 0.30, large ≈ 0.50. (CIs for Wilcoxon median shifts are non-trivial; we report W, p, and r.)

- **Outcomes.** Primary self-reported outcomes: *overall rating*, *emotional engagement*, *robot influenced my mood*, *understood my emotion*, *adapted to my emotion*, *experience change*, and the *signed valence shift*

$$\Delta v = v_{\text{end}} - v_{\text{start}},$$

derived from start/end mood valence. (*Likert*-type items are treated as approximately interval for means and t-tests; Wilcoxon is used when non-normality is evident.)

- **Reporting.** For each outcome we report: n (paired sample size), means by condition, mean difference (*with* − *no*), the statistical test and p-value, and the corresponding effect size (d_z or r). Unless noted, tests are two-tailed with $\alpha = 0.05$, and missing pairs are excluded listwise. If desired, a multiple-comparison control (e.g., BH–FDR) can be applied post hoc to the family of outcomes.

- **Effect size interpretation (heuristics).**

Effect size	**Small**	**Medium**	**Large**
Cohen's d_z (paired)	0.20	0.50	0.80
$r = \frac{Z}{\sqrt{n}}$ (Wilcoxon)	0.10	0.30	0.50

Table 7.6: Rule-of-thumb thresholds; practical significance remains context-dependent.

Paired comparison results (with-data − no-data). Comparisons show significant gains for the adaptive policy on several user-reported outcomes:

- **Emotional engagement** (n=16): no-data = 4.688, with-data = 6.500; diff = +1.812; paired t, p=0.0053; d_z=0.814; 95% CI [+0.63, +3.00].
- **Robot influenced my mood** (n=15): no-data = 6.733, with-data = 7.600; diff = +0.867; paired t, p=0.0267; d_z=0.639; 95% CI [+0.12, +1.62].
- **Experience change** (n=16): no-data = 6.312, with-data = 7.688; diff = +1.375; paired t, p=0.0176; d_z=0.667; 95% CI [+0.28, +2.47].
- **Perceived mood change** (n=15): no-data = 5.800, with-data = 7.533; diff = +1.733; Wilcoxon W, p=0.0145; r≈0.631.

Non-significant outcomes (two-tailed $\alpha = 0.05$):

- **Overall rating** (n=15): no-data = 7.067, with-data = 7.667; diff = +0.600; paired t, p=0.209; d_z=0.340; 95% CI [−0.38, +1.58].
- **Adapted to my emotion** (n=16): no-data = 5.688, with-data = 6.250; diff = +0.562; paired t, p=0.278; d_z=0.281; 95% CI [−0.50, +1.63].
- **Understood my emotion** (n=16): no-data = 5.688, with-data = 6.812; diff = +1.125; paired t, p=0.076; d_z=0.476; 95% CI [−0.13, +2.38] (trend).
- **Δ valence (end–start)** (n=15): no-data = 0.533, with-data = 0.667; diff = +0.133; Wilcoxon W, p=0.593; r≈0.138.

Summary. The adaptive (*with-data*) policy yields higher *emotional engagement*, stronger *perceived influence on mood*, greater *experience change*, and higher *perceived mood change* during interaction, with medium-to-large effect sizes. Other ratings (overall, explicit adaptivity, perceived understanding) and the signed valence shift Δv did not differ significantly in this sample. Overall, these results support the effectiveness of data-driven adaptation in the robot's interaction policy.

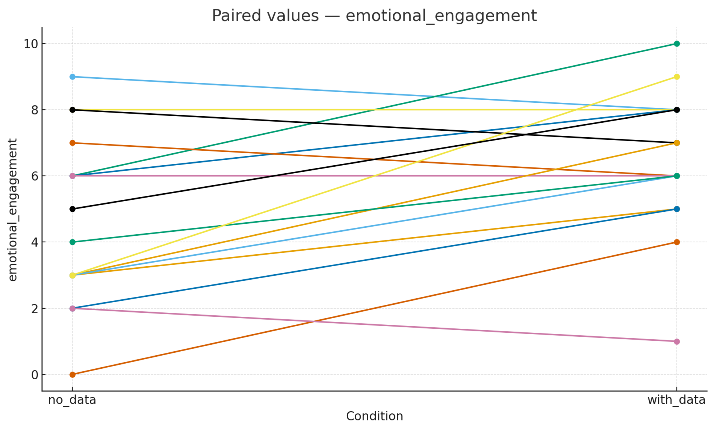

Figure 7.3: Paired values for *Emotional engagement*.

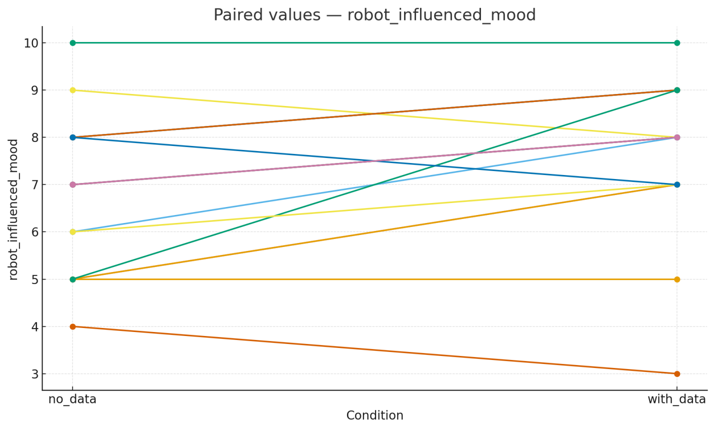

Figure 7.4: Paired values for *Robot influenced my mood*.

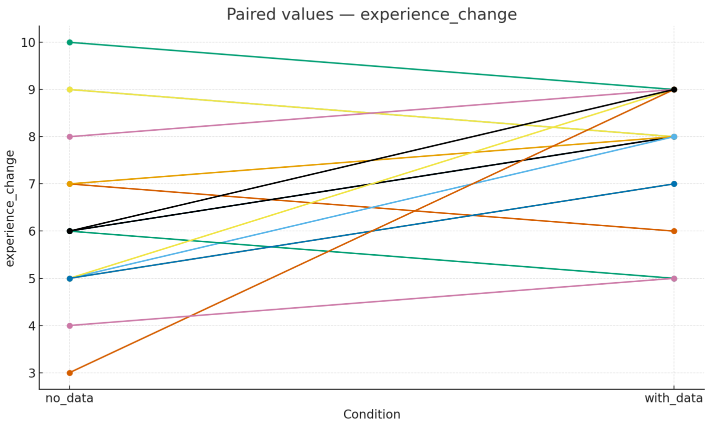

Figure 7.5: Paired values for *Experience change*.

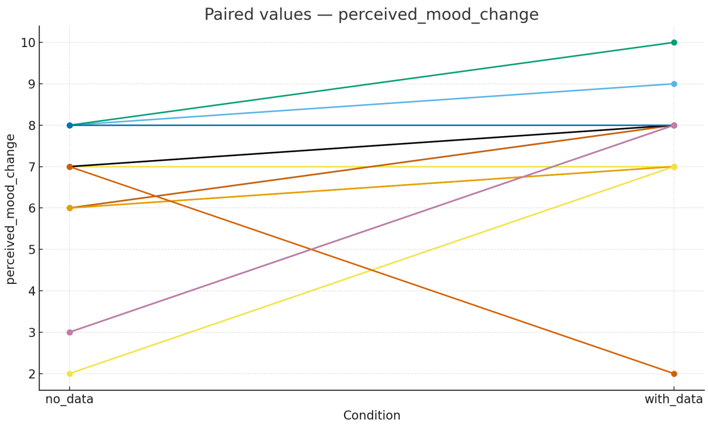

Figure 7.6: Paired values for *Perceived mood change*.

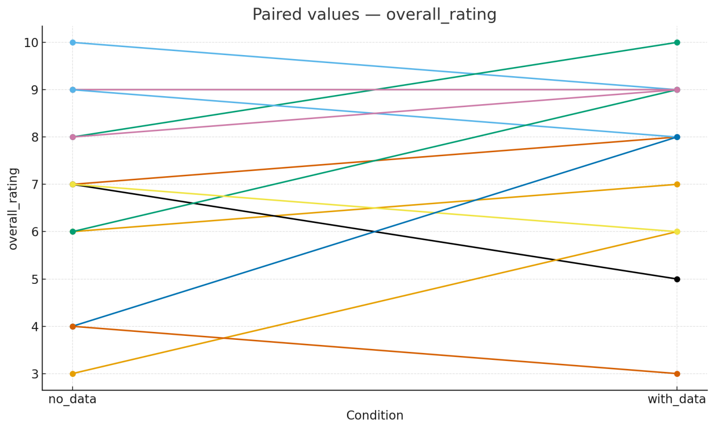

Figure 7.7: Paired values for *Overall rating*.

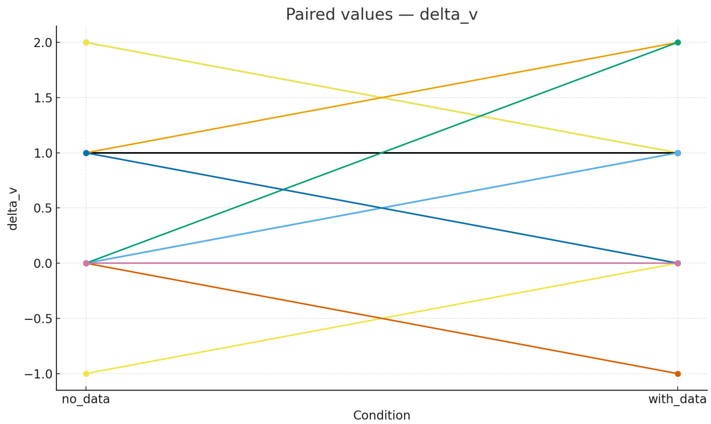

Figure 7.8: Paired values for Δ *valence (end–start)*.

Concluding Remarks

This chapter presented the experimental validation of SUNNY's *mood estimation* and *interaction* capabilities in office-like settings. The project remains an *open and ongoing research effort*: the work conducted here establishes a methodological and technical foundation that can serve as a springboard for future studies, providing both a reproducible pipeline and baseline results upon which larger-scale evaluations and model refinements can be built.

Answers to the Research Questions.

- **RQ1 (Multimodal non-invasive mood/health estimation).** The six-class evaluation (Experiment 1) shows overall accuracy **58.82%** with $\kappa = 0.42$ (moderate agreement), and a more robust **Positive/Neutral/Negative** abstraction with accuracy **73.53%** and $\kappa = 0.594$ (moderate-to-substantial). SUNNY reliably separates *positive* vs. *negative* moods, while *neutral* and low-support negative categories remain challenging. *Conclusion:* RQ1 is **partially supported**: the approach is effective at a coarse-grained level (PNN) using only non-invasive sensing, whereas fine-grained emotions would benefit from more data and calibrated fusion.

- **RQ2 (Effectiveness of robot-led interventions).** The within-subject comparison (Experiment 2) indicates that the *adaptive* (with-data) policy significantly improves *emotional engagement* (p=0.0053, d_z=0.814), *perceived influence on mood* (p=0.0267, d_z=0.639), *experience change* (p=0.0176, d_z=0.667), and *perceived mood change* (Wilcoxon p=0.0145, $r\approx$0.631). Some outcomes (overall rating, explicit adaptivity, perceived understanding) did not reach significance in this sample. *Conclusion:* RQ2 is **supported** on several key user-reported dimensions, indicating that data-driven adaptation yields a measurable benefit.

Scope and limitations. The evidence presented here should be interpreted in light of: (i) limited sample sizes for the paired analysis ($n\approx$15–16 per outcome); (ii) class imbalance and sparse categories in Experiment 1 (36 recorded interactions, labels from >20 observers); (iii) a *rule-based* fusion, chosen for transparency, rather than a learned temporal/multimodal model; (iv) the absence (in this iteration) of the valence–arousal projection and of fully validated clinical thresholds (thermal/mmWave) in the online loop.

Implications. Taken together, the results support SUNNY as a viable *non-invasive* social-robot platform for *real-time, coarse-grained mood awareness* and *adaptive interventions* in office environments. For online interaction, the PNN abstraction appears practically useful; moving to finer emotion labels will likely require larger datasets, improved SER domain adaptation, learned (or Bayesian/fuzzy) fusion with temporal smoothing, and calibration against longitudinal logs.

Outlook and next steps. Future work (see Chapter 8) will prioritize: (i) implementing the Circumplex (valence–arousal) projection alongside discrete labels; (ii) upgrading fusion beyond rules (e.g., Bayesian/Dempster–Shafer, HMM/RNN-based temporal models, or end-to-end multimodal transformers); (iii) threshold validation for thermal/mmWave against ground-truth instrumentation; (iv) dataset growth (more interactions and observers) to tighten CIs, enable stronger statistical power, and tune decision thresholds; (v) policy refinement and safety-by-design (privacy-preserving data minimization, on-device processing) to support long-term deployments.

In summary, the thesis delivers a *research baseline*—hardware, software, and methodology—on which more ambitious, statistically powered studies can iterate, improving both the precision of mood/health estimation and the impact of adaptive, ethical robot interventions in everyday office life.

Chapter 8

Future Developments

8.1 Overview

This chapter outlines *possible* directions to evolve SUNNY beyond the present proof-of-concept. These are *proposals* rather than commitments, meant to inform future work that others or future iterations of the project could pursue. Building on the sensing fusion (Chapter 5.3) and the experimental evidence (Chapter 7), we sketch ideas for: (i) richer affect representations, (ii) calibrated non-invasive health cues, (iii) confidence-aware decisions, (iv) learned/probabilistic fusion, (v) navigation autonomy and operational reliability, and (vi) rigorous evaluation.

8.2 Planned Enhancements

8.2.1 Circumplex Projection (Valence–Arousal)

Status: theoretically specified, not yet implemented. To complement the discrete `mood` label, a continuous projection into the Circumplex Model of Affect (valence v, arousal a) [40, 41] has been designed at the conceptual level. The idea is to define a mapping $\mathcal{M}$ from discrete emotions {Angry, Sad, Happy, Surprise, Fear, Disgust, Contempt, Neutral} to normalized coordinates $(v, a) \in [-1, 1]^2$ (tunable in pilot trials), and to combine FER/SER via confidence–weighted averaging:

$$(v, a)^{\text{raw}} = w_{\text{FER}}\,\mathcal{M}(e_{\text{FER}}) + w_{\text{SER}}\,\mathcal{M}(e_{\text{SER}}), \qquad w_{\text{FER}} + w_{\text{SER}} = 1.$$

Where: e_{FER} and e_{SER} are discrete outputs; $w_{\text{FER}}, w_{\text{SER}} \in [0, 1]$ could reflect signal quality (face detector score, speech VAD/length/confidence). To reduce jitter, one *could* apply an exponential moving average (EMA):

$$(v,a)_t = \alpha\,(v,a)_t^{\text{raw}} + (1-\alpha)\,(v,a)_{t-1}, \quad \alpha \in [0.2, 0.5].$$

If implemented, the VA pair would be logged alongside `mood`/`intensity` for later analysis and smoother interaction.

8.2.2 Health Heuristics (Thermal & mmWave HR)

In keeping with the non-invasive design, thermal and mmWave signals are considered *coarse* health cues rather than diagnostics. For **fever screening**, a practical (future) approach could estimate a stable forehead temperature (e.g., running median on a short window with light morphological filtering) and raise a flag when a configurable threshold θ_{fever} is exceeded, with hysteresis to avoid chattering. A conservative operating band might place θ_{fever} in 37.5–38.0 °C (e.g., 38.0 °C in CDC guidance; 37.5 °C commonly used in Japan during COVID-19), acknowledging the known limitations of thermography for infectious-disease screening [42].

For **heart rate (HR)** from a 60–77 GHz mmWave sensor, a future implementation could median-filter the signal and normalize to a personal baseline:

$$z_{\text{HR}} = \frac{\text{HR} - \mu_{\text{pers}}}{\sigma_{\text{pers}} + \varepsilon}, \qquad s_{\text{HR}} = \text{clip}(z_{\text{HR}}, 0, 1),$$

then gently modulate arousal as $a \leftarrow a + \beta\, s_{\text{HR}}$ with a small gain $\beta \in [0, 0.3]$. Prior studies suggest office-like, short-range FMCW mmWave can recover respiration/HR non-contact [43, 44]; any adoption would require calibration against references (e.g., contact thermometers, pulse-ox/ECG) on a small cohort.

8.2.3 Decision & Confidence

Rather than hard decisions, a *confidence score* $C \in [0, 1]$ could combine (i) FER/SER agreement, (ii) modality quality (FER score, SER VAD/length, HR availability), and (iii) temporal stability (EMA variance). Low C or unstable/missing health cues would trigger conservative policies (neutral prompts, avoidance of sensitive suggestions).

8.2.4 Fusion Beyond Rules

If more longitudinal data become available, the current transparent, rule-based fusion (Chapter 5.3) could be extended with:

- Bayesian or Dempster–Shafer weighting for uncertainty-aware decision fusion;
- Temporal models (HMM/CRF, or lightweight RNN/GRU) to exploit sequential coherence in (v, a) and discrete moods;
- Small neural aggregators using FER/SER logits, quality features, and mmWave/thermal cues to predict mood and (v, a) with calibration constraints.

8.2.5 Navigation & Operational Autonomy

Beyond static-map navigation with real-time obstacle avoidance, future work *could* explore:

- **Auto-docking** on a fast-charge base to mitigate the current $\sim$2 h battery limit;
- **Semantic waypoints** (e.g., meeting rooms, cafeteria) exposed via safe-goal APIs tied to map markers;
- **Lightweight map refresh** (periodic static-layer updates) to accommodate minor layout changes without full SLAM.

8.2.6 Evaluation & Statistical Power

Subsequent studies *could* target larger samples and repeated-measures designs with counterbalancing, pre-registered hypotheses, and multiple-comparison control (e.g., BH–FDR). Power analyses based on current effects (d_z, r) would inform target n and help tighten confidence intervals; observer pools could be expanded to reduce uncertainty for minority classes.

8.3 Summary

The items above are intentionally framed as *possibilities*. The Circumplex projection has been theoretically outlined but not implemented; the remaining points are prospective ideas that may guide future iterations. Taken together, they indicate how SUNNY might evolve toward more adaptive, calibrated, and privacy-preserving HRI in office environments, should resources and scope allow.

Acknowledgements

This work exists because many generous people shared their time, ideas, and faith in me when I needed it most.

First and foremost, I am deeply grateful to my supervisors, Prof. Bruno Siciliano and Prof. Yasuhisa Hirata. Your guidance, your patience with my questions, and the trust you placed in this project shaped both the research and the person behind it. Thank you for offering me the opportunity to carry out my thesis work in Japan, a formative experience both scientifically and personally, and for encouraging (and supporting) me to deliver early, tangible milestones on the robot, which made it possible to bring a live demo to ICRA 2024 in Yokohama.

I owe a special debt of gratitude to my co-supervisors, Prof. Ankit Ravankar and Prof. Jose V. Salazar L., whose thoughtful critiques and timely advice consistently pushed the thesis toward greater clarity and rigor.

My sincere thanks go to Hanada-San and all the other colleagues at the SRD Lab (Tohoku University). Thank you for the hallway conversations that solved stubborn bugs, for sharing equipment and space, and for welcoming a robot that insisted on taking the long way around the office. To all participants who lent their time and presence to the experiments: your patience gave this work its meaning.

Finally, to everyone who taught me that good research is also an act of care, toward the people we hope to help and the colleagues we walk with, thank you.

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